$$C_{eH}^{i1\_i2\_} \to \frac{1}{8} s_{4\gamma} s_{\gamma} \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (g_1^2 + g_2^2) +$$

$$\frac{1}{16\pi^2} (\frac{1}{384} \frac{1}{m_{\Phi}^{2}} (s_{4\gamma} y_e^{i1i2} (6 s_{4\gamma} c_{\gamma} (g_1^2+g_2^2)^2 + s_{\gamma} (3 g_1^4 (-1 + 6 c_{\gamma}^2) +$$

$$24 g_1^2 g_2^2 (1 + 3 c_{\gamma}^2) + g_2^4 (-5 + 54 c_{\gamma}^2)) + 18 s_{\gamma}^3 (g_1^2+g_2^2)(g_1^2 + 3 g_2^2)) +$$

$$6 s_{\gamma} \overline{y_e}^{pr} (y_e^{i1r} (-s_{4\gamma} y_e^{pi2} (g_1^2+g_2^2)(10 c_{\gamma}^2 - 3(1+s_{\gamma}^2)) - 16 s_{\gamma} c_{\gamma}^3 \overline{y_e}^{st} y_e^{pt} y_e^{si2}) -$$

$$8 s_{\gamma} c_{\gamma} y_e^{i1i2} (c_{2\gamma}^2 y_e^{pr} (g_1^2+g_2^2) - 8 c_{\gamma}^2 \overline{y_e}^{st} y_e^{pt} y_e^{sr})))) +$$

$$\frac{1}{9} c_{\gamma} g_2^4 y_e^{i1i2} \mathrm{LF}_{3,0}[m_2] + \frac{1}{6} c_{\gamma} g_2^4 y_e^{i1i2} \mathrm{LF}_{4,-1}[m_2] - \frac{8}{45} c_{\gamma} g_2^4 y_e^{i1i2} \mathrm{LF}_{5,-2}[m_2] +$$

$$\frac{1}{16} \sum_p g_1^2 \frac{1}{m_{\Phi}^{4}} y_e^{i1i2} (g_1^2+g_2^2)$$

$$(s_{2\gamma} s_{4\gamma} c_{\gamma} + s_{\gamma} (2 s_{2\gamma} (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2) + s_{4\gamma} c_{2\gamma})) \mathrm{LF}_{1,0}[m_{\tilde{d}}^{p}] -$$

$$\frac{1}{12} \sum_p s_{2\gamma} s_{\gamma} c_{2\gamma} g_1^4 \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} \mathrm{LF}_{2,0}[m_{\tilde{d}}^{p}] - \frac{3}{16} \frac{1}{m_{\Phi}^{4}} \overline{y_d}^{pr} y_d^{pr} y_e^{i1i2} (g_1^2+g_2^2)$$

$$(s_{2\gamma} s_{4\gamma} c_{\gamma} + 2 s_{2\gamma} s_{\gamma} (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2) - 2 s_{4\gamma} s_{\gamma}^3) \mathrm{LF}_{1,0}[m_{\tilde{d}}^{r}] +$$

$$\frac{1}{4} s_{2\gamma} s_{\gamma} g_1^2 \frac{1}{m_{\Phi}^{2}} \overline{y_d}^{pr} y_d^{pr} y_e^{i1i2} (c_{2\gamma} + 2 c_{\gamma}^2) \mathrm{LF}_{2,0}[m_{\tilde{d}}^{r}] + \frac{1}{16} \sum_p g_1^2 \frac{1}{m_{\Phi}^{4}} y_e^{i1i2}$$

$$(g_1^2+g_2^2) (s_{2\gamma} s_{4\gamma} c_{\gamma} + s_{\gamma} (2 s_{2\gamma} (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2) + s_{4\gamma} c_{2\gamma})) \mathrm{LF}_{1,0}[m_{\tilde{e}}^{p}] -$$

$$\frac{1}{4} \sum_p s_{2\gamma} s_{\gamma} c_{2\gamma} g_1^4 \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} \mathrm{LF}_{2,0}[m_{\tilde{e}}^{p}] - \frac{1}{16} \frac{1}{m_{\Phi}^{4}} \overline{y_e}^{pr} y_e^{pr} y_e^{i1i2} (g_1^2+g_2^2)$$

$$(s_{2\gamma} s_{4\gamma} c_{\gamma} + 2 s_{2\gamma} s_{\gamma} (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2) - 2 s_{4\gamma} s_{\gamma}^3) \mathrm{LF}_{1,0}[m_{\tilde{e}}^{r}] +$$

$$\frac{1}{4} s_{2\gamma} s_{\gamma} g_1^2 \frac{1}{m_{\Phi}^{2}} \overline{y_e}^{pr} y_e^{pr} y_e^{i1i2} (c_{2\gamma} + 2 c_{\gamma}^2) \mathrm{LF}_{2,0}[m_{\tilde{e}}^{r}] +$$

$$\frac{1}{16} \frac{1}{m_{\Phi}^{4}} y_e^{i1i2} (g_1^2+g_2^2) (-\overline{y_e}^{pr} y_e^{pr} (s_{2\gamma} s_{4\gamma} c_{\gamma} + 2 s_{2\gamma} s_{\gamma} (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2) - 2 s_{4\gamma} s_{\gamma}^3) -$$

$$\sum_p g_1^2 (s_{2\gamma} s_{4\gamma} c_{\gamma} + s_{\gamma} (2 s_{2\gamma} (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2) + s_{4\gamma} c_{2\gamma}))) \mathrm{LF}_{1,0}[m_{\tilde{l}}^{p}] -$$

$$\frac{1}{8} s_{2\gamma} s_{\gamma} \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (\overline{y_e}^{pr} y_e^{pr} (c_{2\gamma} + 2 c_{\gamma}^2)(g_1^2 - g_2^2) + \sum_p c_{2\gamma} (g_1^4 + g_2^4)) \mathrm{LF}_{2,0}[m_{\tilde{l}}^{p}] +$$

$$\frac{1}{72} c_{\gamma} g_2^2 y_e^{i1i2} (36 c_{2\gamma} c_{\gamma}^2 \overline{y_e}^{pr} y_e^{pr} + \sum_p g_2^2 (4 - 9 c_{2\gamma}^2)) \mathrm{LF}_{3,0}[m_{\tilde{l}}^{p}] +$$

$$\frac{1}{48} c_{\gamma} g_2^2 y_e^{i1i2} (-24 c_{2\gamma} c_{\gamma}^2 \overline{y_e}^{pr} y_e^{pr} + \sum_p g_2^2 (-5 + 6 c_{2\gamma}^2)) \mathrm{LF}_{4,-1}[m_{\tilde{l}}^{p}] +$$

$$\frac{2}{45} \sum_p c_{\gamma} g_2^4 y_e^{i1i2} \mathrm{LF}_{5,-2}[m_{\tilde{l}}^{p}] +$$

$$\frac{1}{16} \frac{1}{m_{\Phi}^{4}} y_e^{i1i2} (g_1^2+g_2^2) (-3 \overline{y_d}^{pr} y_d^{pr} (s_{2\gamma} s_{4\gamma} c_{\gamma} + 2 s_{2\gamma} s_{\gamma} (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2) - 2 s_{4\gamma} s_{\gamma}^3) +$$

$$3 \overline{y_u}^{pr} y_u^{pr} (s_{2\gamma} s_{4\gamma} c_{\gamma} + 2 s_{\gamma} (s_{2\gamma} (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2) + s_{4\gamma} c_{\gamma}^2)) +$$

$$\sum_p g_1^2 (s_{2\gamma} s_{4\gamma} c_{\gamma} + s_{\gamma} (2 s_{2\gamma} (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2) + s_{4\gamma} c_{2\gamma}))) \mathrm{LF}_{1,0}[m_{\tilde{q}}^{p}] +$$

$$\frac{1}{24} s_{2\gamma} s_{\gamma} \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (3 \overline{y_d}^{pr} y_d^{pr} (c_{2\gamma} + 2 c_{\gamma}^2)(g_1^2 + 3 g_2^2) -$$

$$3 \overline{y_u}^{pr} y_u^{pr} (g_1^2 - 3 g_2^2)(c_{2\gamma} - 2 s_{\gamma}^2) - \sum_p c_{2\gamma} (g_1^4 + 9 g_2^4)) \mathrm{LF}_{2,0}[m_{\tilde{q}}^{p}] +$$

$$\frac{1}{24} c_{\gamma} g_2^2 y_e^{i1i2} (36 c_{2\gamma} (c_{\gamma}^2 \overline{y_d}^{pr} y_d^{pr} - s_{\gamma}^2 \overline{y_u}^{pr} y_u^{pr}) + \sum_p g_2^2 (4 - 9 c_{2\gamma}^2)) \mathrm{LF}_{3,0}[m_{\tilde{q}}^{p}] +$$

$$\frac{1}{16} c_{\gamma} g_2^2 y_e^{i1i2} (24 c_{2\gamma} (-c_{\gamma}^2 \overline{y_d}^{pr} y_d^{pr} + s_{\gamma}^2 \overline{y_u}^{pr} y_u^{pr}) + \sum_p g_2^2 (-5 + 6 c_{2\gamma}^2)) \mathrm{LF}_{4,-1}[m_{\tilde{q}}^{p}] +$$

$$\frac{2}{15} \sum_p c_{\gamma} g_2^4 y_e^{i1i2} \mathrm{LF}_{5,-2}[m_{\tilde{q}}^{p}] - \frac{1}{8} \sum_p g_1^2 \frac{1}{m_{\Phi}^{4}} y_e^{i1i2} (g_1^2+g_2^2)$$

$$(s_{2\gamma} s_{4\gamma} c_{\gamma} + s_{\gamma} (2 s_{2\gamma} (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2) + s_{4\gamma} c_{2\gamma})) \mathrm{LF}_{1,0}[m_{\tilde{u}}^{p}] -$$

$$\frac{1}{3} \sum_p s_{2\gamma} s_{\gamma} c_{2\gamma} g_1^4 \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} \mathrm{LF}_{2,0}[m_{\tilde{u}}^{p}] + \frac{3}{16} \frac{1}{m_{\Phi}^{4}} y_e^{i1i2} \overline{y_u}^{pr} y_u^{pr} (g_1^2+g_2^2)$$

$$(s_{2\gamma} s_{4\gamma} c_{\gamma} + 2 s_{\gamma} (s_{2\gamma} (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2) + s_{4\gamma} c_{\gamma}^2)) \mathrm{LF}_{1,0}[m_{\tilde{u}}^{r}] +$$

$$\frac{1}{2} s_{2\gamma} s_{\gamma} g_1^2 \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} \overline{y_u}^{pr} y_u^{pr} (c_{2\gamma} - 2 s_{\gamma}^2) \mathrm{LF}_{2,0}[m_{\tilde{u}}^{r}] +$$

$$\frac{1}{64} s_{4\gamma} \frac{1}{m_{\Phi}^{4}} y_e^{i1i2} (g_1^2+g_2^2) (3 s_{4\gamma} c_{\gamma} (g_1^2+g_2^2) +$$

$$2 s_{\gamma} (g_1^2 (-1 + 6 c_{2\gamma}^2 + 3 c_{4\gamma} - 3 s_{2\gamma}^2) + 3 g_2^2 (-1 + 2 c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2)))$$

$$\mathrm{LF}_{1,0}[m_{\Phi}] + \frac{1}{64} s_{4\gamma} s_{\gamma} \frac{1}{m_{\Phi}^{2}} (g_1^2+g_2^2) (-6 s_{\gamma}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} +$$

$$y_e^{i1i2} (g_1^2 (3 + 7 c_{4\gamma} - 10 s_{2\gamma}^2) + g_2^2 (15 + 7 c_{4\gamma} - 10 s_{2\gamma}^2))) \mathrm{LF}_{1,1}[m_{\Phi}] -$$

$$\frac{1}{32} s_{4\gamma} s_{\gamma} (g_1^2+g_2^2) (-18 c_{\gamma}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} + y_e^{i1i2} (g_1^2 + 3 g_2^2)) \mathrm{LF}_{1,2}[m_{\Phi}] -$$

$$\frac{1}{64} s_{4\gamma} s_{\gamma} \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (g_1^2+g_2^2) (g_2^2 (-3 + 5 c_{4\gamma} - 2 s_{2\gamma}^2) + g_1^2 (1 + 5 c_{4\gamma} - 2 s_{2\gamma}^2)) \mathrm{LF}_{2,0}[m_{\Phi}] -$$

$$\frac{1}{32} c_{\gamma} s_{\gamma}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} (g_2^2 (-3 + 5 c_{4\gamma} - 8 s_{2\gamma}^2) + g_1^2 (1 + 5 c_{4\gamma} - 8 s_{2\gamma}^2)) \mathrm{LF}_{2,1}[m_{\Phi}] -$$

$$\frac{1}{1152} c_{\gamma} y_e^{i1i2} (9 g_1^4 ((-1 + c_{4\gamma})^2 + 4 s_{2\gamma}^4) + 18 g_1^2 g_2^2 (-3 + c_{4\gamma} (2 + c_{4\gamma}) + 4 s_{2\gamma}^4) +$$

$$g_2^4 (17 + 9 c_{4\gamma} (6 + c_{4\gamma}) + 36 s_{2\gamma}^4)) \mathrm{LF}_{3,0}[m_{\Phi}] +$$

$$\frac{1}{384} c_{\gamma} y_e^{i1i2} (3 g_1^4 ((-1 + c_{4\gamma})^2 + 4 s_{2\gamma}^4) + 6 g_1^2 g_2^2 (-3 + c_{4\gamma} (2 + c_{4\gamma}) + 4 s_{2\gamma}^4) +$$

$$g_2^4 (-13 + 3 c_{4\gamma} (6 + c_{4\gamma}) + 12 s_{2\gamma}^4)) \mathrm{LF}_{4,-1}[m_{\Phi}] + \frac{2}{45} c_{\gamma} g_2^4 y_e^{i1i2} \mathrm{LF}_{5,-2}[m_{\Phi}] +$$

$$\frac{1}{18} c_{\gamma} g_2^4 y_e^{i1i2} \mathrm{LF}_{3,0}[\tilde{\mu}] + \frac{1}{12} c_{\gamma} g_2^4 y_e^{i1i2} \mathrm{LF}_{4,-1}[\tilde{\mu}] - \frac{4}{45} c_{\gamma} g_2^4 y_e^{i1i2} \mathrm{LF}_{5,-2}[\tilde{\mu}] -$$

$$g_1^2 c_{\gamma}^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,0}[m_1, m_{\tilde{e}}^{r}] + 2 g_1^2 c_{\gamma}^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,-1}[m_1, m_{\tilde{e}}^{r}] -$$

$$g_1^2 c_{\gamma}^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{4,1,-2}[m_1, m_{\tilde{e}}^{r}] - \frac{1}{8} s_{4\gamma} s_{\gamma} g_1^2 \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (g_1^2+g_2^2) \mathrm{LF}_{1,1,0}[m_1, m_{\tilde{e}}^{i2}] +$$

$$\frac{1}{16} s_{4\gamma} s_{\gamma} g_1^2 \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (g_1^2+g_2^2) \mathrm{LF}_{2,1,-1}[m_1, m_{\tilde{e}}^{i2}] +$$

$$\frac{1}{4} c_{2\gamma} c_{\gamma} g_1^4 y_e^{i1i2} \mathrm{LF}_{2,2,-1}[m_1, m_{\tilde{e}}^{i2}] - \frac{1}{4} g_1^2 c_{\gamma}^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,0}[m_1, m_{\tilde{l}}^{p}] +$$

$$\frac{1}{2} g_1^2 c_{\gamma}^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,-1}[m_1, m_{\tilde{l}}^{p}] - \frac{1}{4} g_1^2 c_{\gamma}^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{4,1,-2}[m_1, m_{\tilde{l}}^{p}] -$$

$$\frac{1}{32} s_{4\gamma} s_{\gamma} g_1^2 \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (g_1^2+g_2^2) \mathrm{LF}_{1,1,0}[m_1, m_{\tilde{l}}^{i1}] +$$

$$\frac{1}{64} s_{4\gamma} s_{\gamma} g_1^2 \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (g_1^2+g_2^2) \mathrm{LF}_{2,1,-1}[m_1, m_{\tilde{l}}^{i1}] +$$

$$\frac{1}{32} c_{2\gamma} c_{\gamma} g_1^2 y_e^{i1i2} (-g_1^2 + g_2^2) \mathrm{LF}_{2,2,-1}[m_1, m_{\tilde{l}}^{i1}] -$$

$$\frac{1}{8} s_{4\gamma} s_{\gamma} g_1^2 \frac{1}{m_{\Phi}^{4}} y_e^{i1i2} (g_1^2+g_2^2) \mathrm{LF}_{1,1,-1}[m_1, \tilde{\mu}] - \frac{1}{8} m_1 \tilde{\mu} g_1^2 \frac{1}{m_{\Phi}^{4}} y_e^{i1i2} (g_1^2+g_2^2)$$

$$(s_{4\gamma} c_{\gamma}^3 + 2 s_{\gamma} c_{\gamma}^2 (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2) - 3 s_{4\gamma} c_{\gamma} s_{\gamma}^2 - 2 s_{\gamma}^3 (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2))$$

$$\mathrm{LF}_{1,1,0}[m_1, \tilde{\mu}] + s_{\gamma} \tilde{\mu} g_1^4 \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (-c_{\gamma}^2 + s_{\gamma}^2)(m_1 + s_{\gamma} \tilde{\mu} c_{\gamma}) \mathrm{LF}_{2,2,-1}[m_1, \tilde{\mu}] +$$

$$c_{\gamma} g_1^4 m_1^2 \frac{1}{m_{\Phi}^{2}} s_{\gamma}^2 \tilde{\mu}^2 y_e^{i1i2} (-c_{\gamma}^2 + s_{\gamma}^2) \mathrm{LF}_{2,2,0}[m_1, \tilde{\mu}] -$$

$$\frac{3}{4} g_2^2 c_{\gamma}^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,0}[m_2, m_{\tilde{l}}^{p}] + \frac{3}{2} g_2^2 c_{\gamma}^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,-1}[m_2, m_{\tilde{l}}^{p}] -$$

$$\frac{3}{4} g_2^2 c_{\gamma}^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{4,1,-2}[m_2, m_{\tilde{l}}^{p}] -$$

$$\frac{3}{32} s_{4\gamma} s_{\gamma} g_2^2 \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (g_1^2+g_2^2) \mathrm{LF}_{1,1,0}[m_2, m_{\tilde{l}}^{i1}] +$$

$$\frac{3}{64} s_{4\gamma} s_{\gamma} g_2^2 \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (g_1^2+g_2^2) \mathrm{LF}_{2,1,-1}[m_2, m_{\tilde{l}}^{i1}] -$$

$$\frac{1}{32} c_{2\gamma} c_{\gamma} g_2^2 y_e^{i1i2} (3 g_1^2 + g_2^2) \mathrm{LF}_{2,2,-1}[m_2, m_{\tilde{l}}^{i1}] -$$

$$\frac{3}{8} s_{4\gamma} s_{\gamma} g_2^2 \frac{1}{m_{\Phi}^{4}} y_e^{i1i2} (g_1^2+g_2^2) \mathrm{LF}_{1,1,-1}[m_2, \tilde{\mu}] - \frac{3}{8} m_2 \tilde{\mu} g_2^2 \frac{1}{m_{\Phi}^{4}} y_e^{i1i2} (g_1^2+g_2^2)$$

$$(s_{4\gamma} c_{\gamma}^3 + 2 s_{\gamma} c_{\gamma}^2 (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2) - 3 s_{4\gamma} c_{\gamma} s_{\gamma}^2 - 2 s_{\gamma}^3 (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2))$$

$$\mathrm{LF}_{1,1,0}[m_2, \tilde{\mu}] - \frac{2}{3} c_{\gamma} g_2^4 y_e^{i1i2} \mathrm{LF}_{2,1,0}[m_2, \tilde{\mu}] +$$

$$2 c_{\gamma} g_2^4 \frac{1}{m_{\Phi}^{2}} s_{\gamma}^2 y_e^{i1i2} (c_{\gamma}^2 - s_{\gamma}^2) \mathrm{LF}_{2,2,-2}[m_2, \tilde{\mu}] - \frac{1}{12} g_2^4 \frac{1}{m_{\Phi}^{2}} y_e^{i1i2}$$

$$(c_{\gamma} (m_{\Phi}^2 c_{\gamma}^2 (8 + 9 c_{\gamma}^2) + 8 s_{\gamma}^2 (m_{\Phi}^2 + 3 c_{\gamma}^2 (m_2^2 - m_{\Phi}^2)) + 3 s_{\gamma}^4 (-8 m_2^2 + 3 m_{\Phi}^2)) +$$

$$36 m_2 s_{\gamma} \tilde{\mu} (c_{\gamma}^4 - s_{\gamma}^4) + 36 c_{\gamma} s_{\gamma}^2 \tilde{\mu}^2 (c_{\gamma}^2 - s_{\gamma}^2)) \mathrm{LF}_{2,2,-1}[m_2, \tilde{\mu}] -$$

$$\frac{1}{3} m_2 s_{\gamma} c_{\gamma} g_2^4 \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (4 \tilde{\mu} c_{\gamma} m_{\Phi}^2 - 9 m_2 s_{\gamma}^3 \tilde{\mu}^2 + 3 m_2 s_{\gamma} c_{\gamma}^2 (m_{\Phi}^2 + 3 \tilde{\mu}^2)) \mathrm{LF}_{2,2,0}[m_2, \tilde{\mu}] +$$

$$\frac{1}{3} c_{\gamma} g_2^4 y_e^{i1i2} \mathrm{LF}_{3,1,-1}[m_2, \tilde{\mu}] +$$

$$\frac{1}{3} c_{\gamma} g_2^4 y_e^{i1i2} (6 c_{\gamma}^4 + s_{\gamma}^2 + 6 s_{\gamma}^4 + c_{\gamma}^2 (1 - 12 s_{\gamma}^2)) \mathrm{LF}_{3,2,-2}[m_2, \tilde{\mu}] +$$

$$\frac{1}{6} m_2 c_{\gamma} g_2^4 y_e^{i1i2} (-9 m_2 (c_{\gamma}^2 - s_{\gamma}^2)^2 + 4 s_{\gamma} \tilde{\mu} c_{\gamma}) \mathrm{LF}_{3,2,-1}[m_2, \tilde{\mu}] -$$

$$c_{\gamma} g_2^4 y_e^{i1i2} \mathrm{LF}_{3,3,-3}[m_2, \tilde{\mu}] (c_{\gamma}^2 - s_{\gamma}^2)^2 + c_{\gamma} g_2^4 m_2^2 y_e^{i1i2} \mathrm{LF}_{3,3,-2}[m_2, \tilde{\mu}] (c_{\gamma}^2 - s_{\gamma}^2)^2 -$$

$$c_{\gamma} g_2^4 y_e^{i1i2} \mathrm{LF}_{4,2,-3}[m_2, \tilde{\mu}] (c_{\gamma}^2 - s_{\gamma}^2)^2 + c_{\gamma} g_2^4 m_2^2 y_e^{i1i2} \mathrm{LF}_{4,2,-2}[m_2, \tilde{\mu}] (c_{\gamma}^2 - s_{\gamma}^2)^2 -$$

$$\frac{3}{2} s_{2\gamma} s_{\gamma} \frac{1}{m_{\Phi}^{2}} c_{\gamma}^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_e^{i1i2} \mathrm{LF}_{1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}] -$$

$$\frac{3}{8} \frac{1}{m_{\Phi}^{4}} y_e^{i1i2} (g_1^2+g_2^2) (2 s_{\gamma} \tilde{\mu} \overline{y_d}^{pr} (a_d^{pr} (c_{\gamma}^2 (c_{2\gamma}^2 + c_{4\gamma}) - s_{4\gamma} s_{\gamma} c_{\gamma} + s_{2\gamma}^2 s_{\gamma}^2) -$$

$$\tilde{\mu} c_{\gamma} y_d^{pr} (s_{4\gamma} c_{\gamma} + s_{\gamma} (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2))) +$$

$$\overline{a_d}^{pr} (s_{\gamma} a_d^{pr} (s_{4\gamma} c_{\gamma}^2 + 2 s_{\gamma} c_{\gamma} (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2) - s_{4\gamma} s_{\gamma}^2) +$$

$$\tilde{\mu} y_d^{pr} (s_{4\gamma} c_{\gamma}^3 - 2 s_{\gamma} s_{2\gamma}^2 c_{\gamma}^2 - s_{4\gamma} c_{\gamma} s_{\gamma}^2 - 2 s_{\gamma}^3 (c_{2\gamma}^2 + c_{4\gamma})))) \mathrm{LF}_{1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] +$$

$$\frac{1}{2} s_{\gamma} g_1^2 \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (\overline{a_d}^{pr} (s_{2\gamma} c_{\gamma} + s_{\gamma} c_{2\gamma}) + \tilde{\mu} \overline{y_d}^{pr} (c_{2\gamma} c_{\gamma} - s_{2\gamma} s_{\gamma}))$$

$$(c_{\gamma} a_d^{pr} - s_{\gamma} \tilde{\mu} y_d^{pr}) \mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] +$$

$$\frac{1}{4} c_{2\gamma} c_{\gamma} g_1^2 y_e^{i1i2} (c_{\gamma} \overline{a_d}^{pr} - s_{\gamma} \tilde{\mu} \overline{y_d}^{pr})(c_{\gamma} a_d^{pr} - s_{\gamma} \tilde{\mu} y_d^{pr}) \mathrm{LF}_{2,2,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] -$$

$$\frac{1}{2} s_{2\gamma} s_{\gamma} \frac{1}{m_{\Phi}^{2}} c_{\gamma}^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_e^{i1i2} \mathrm{LF}_{1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}] -$$

$$\frac{1}{8} \frac{1}{m_{\Phi}^{4}} y_e^{i1i2} (g_1^2+g_2^2) (2 s_{\gamma} \tilde{\mu} \overline{y_e}^{pr} (a_e^{pr} (c_{\gamma}^2 (c_{2\gamma}^2 + c_{4\gamma}) - s_{4\gamma} s_{\gamma} c_{\gamma} + s_{2\gamma}^2 s_{\gamma}^2) -$$

$$\tilde{\mu} c_{\gamma} y_e^{pr} (s_{4\gamma} c_{\gamma} + s_{\gamma} (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2))) +$$

$$\overline{a_e}^{pr} (s_{\gamma} a_e^{pr} (s_{4\gamma} c_{\gamma}^2 + 2 s_{\gamma} c_{\gamma} (c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2) - s_{4\gamma} s_{\gamma}^2) +$$

$$\tilde{\mu} y_e^{pr} (s_{4\gamma} c_{\gamma}^3 - 2 s_{\gamma} s_{2\gamma}^2 c_{\gamma}^2 - s_{4\gamma} c_{\gamma} s_{\gamma}^2 - 2 s_{\gamma}^3 (c_{2\gamma}^2 + c_{4\gamma})))) \mathrm{LF}_{1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] +$$

$$\frac{1}{2} s_{\gamma} g_1^2 \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (\overline{a_e}^{pr} (s_{2\gamma} c_{\gamma} + s_{\gamma} c_{2\gamma}) + \tilde{\mu} \overline{y_e}^{pr} (c_{2\gamma} c_{\gamma} - s_{2\gamma} s_{\gamma}))$$

$$(c_{\gamma} a_e^{pr} - s_{\gamma} \tilde{\mu} y_e^{pr}) \mathrm{LF}_{2,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] +$$

$$\frac{1}{4} c_{2\gamma} c_{\gamma} g_1^2 y_e^{i1i2} (c_{\gamma} \overline{a_e}^{pr} - s_{\gamma} \tilde{\mu} \overline{y_e}^{pr})(c_{\gamma} a_e^{pr} - s_{\gamma} \tilde{\mu} y_e^{pr}) \mathrm{LF}_{2,2,0}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] -$$

$$\frac{1}{16} s_{4\gamma} s_{\gamma} \frac{1}{m_{\Phi}^{2}} \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} (g_1^2+g_2^2) \mathrm{LF}_{1,1,0}[m_{\tilde{e}}^{r}, \tilde{\mu}] -$$

$$\frac{1}{4} c_{2\gamma} c_{\gamma} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,0}[m_{\tilde{e}}^{r}, \tilde{\mu}] +$$

$$\frac{1}{8} c_{2\gamma} c_{\gamma} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,2,-1}[m_{\tilde{e}}^{r}, \tilde{\mu}] - \frac{1}{2} c_{2\gamma} c_{\gamma} g_1^4 y_e^{i1i2} \mathrm{LF}_{2,1,0}[m_{\tilde{e}}^{i2}, m_1] +$$

$$\frac{1}{16} \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (s_{\gamma} \tilde{\mu} \overline{y_e}^{pr} (c_{\gamma} a_e^{pr} (4 c_{2\gamma} c_{\gamma} (g_1^2 (-1 + c_{2\gamma}) + g_2^2 (1 + c_{2\gamma})) +$$

$$s_{\gamma} (4 s_{2\gamma} (g_1^2 - g_2^2) - 5 s_{4\gamma} (g_1^2+g_2^2))) - \tilde{\mu} y_e^{pr} (2 s_{4\gamma} c_{\gamma}^2 (g_1^2+g_2^2) + 4 s_{\gamma} c_{2\gamma}$$

$$c_{\gamma} (g_1^2 (-1 + c_{2\gamma}) + g_2^2 (1 + c_{2\gamma})) + s_{\gamma}^2 (4 s_{2\gamma} (g_1^2 - g_2^2) - 3 s_{4\gamma} (g_1^2+g_2^2)))) +$$

$$\overline{a_e}^{pr} (s_{\gamma} c_{\gamma} a_e^{pr} (c_{\gamma} (4 s_{2\gamma} (-g_1^2 + g_2^2) + 5 s_{4\gamma} (g_1^2+g_2^2)) +$$

$$4 s_{\gamma} c_{2\gamma} (g_1^2 (-1 + c_{2\gamma}) + g_2^2 (1 + c_{2\gamma}))) +$$

$$\tilde{\mu} y_e^{pr} (2 s_{4\gamma} c_{\gamma}^3 (g_1^2+g_2^2) + c_{\gamma} s_{\gamma}^2 (4 s_{2\gamma} (g_1^2 - g_2^2) - 3 s_{4\gamma} (g_1^2+g_2^2)) - 4 c_{2\gamma} s_{\gamma}^3$$

$$(g_1^2 (-1 + c_{2\gamma}) + g_2^2 (1 + c_{2\gamma}))))) \mathrm{LF}_{2,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + \frac{1}{16} \frac{1}{m_{\Phi}^{2}} y_e^{i1i2} (g_1^2+g_2^2)$$

$$(s_{\gamma} \tilde{\mu} \overline{y_e}^{pr} (c_{\gamma} a_e^{pr} (-4 c_{\gamma} c_{2\gamma}^2 + 5 s_{4\gamma} s_{\gamma}) + \tilde{\mu} y_e^{pr} (2 s_{4\gamma} c_{\gamma}^2 + 4 s_{\gamma} c_{\gamma} c_{2\gamma}^2 - 3 s_{4\gamma} s_{\gamma}^2)) +$$

$$\overline{a_e}^{pr} (-s_{\gamma} c_{\gamma} a_e^{pr} (5 s_{4\gamma} c_{\gamma} + 4 s_{\gamma} c_{2\gamma}^2) + \tilde{\mu} y_e^{pr} (-2 s_{4\gamma} c_{\gamma}^3 + 3 s_{4\gamma} c_{\gamma} s_{\gamma}^2 + 4 s_{\gamma}^3 c_{2\gamma}^2)))$$

$$\mathrm{LF}_{3,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + \frac{1}{4} c_{2\gamma} c_{\gamma} y_e^{i1i2} (g_1^2 (-1 + c_{2\gamma}) + g_2^2 (1 + c_{2\gamma}))$$

$$(c_{\gamma} \overline{a_e}^{pr} - s_{\gamma} \tilde{\mu} \overline{y_e}^{pr})(c_{\gamma} a_e^{pr} - s_{\gamma} \tilde{\mu} y_e^{pr}) \mathrm{LF}_{3,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] -$$

$$\frac{1}{4} c_{2\gamma} c_{\gamma} g_1^2 y_e^{i1i2} (c_{\gamma} \overline{a_e}^{pr} - s_{\gamma} \tilde{\mu} \overline{y_e}^{pr})(c_{\gamma} a_e^{pr} - s_{\gamma} \tilde{\mu} y_e^{pr}) \mathrm{LF}_{3,2,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] -$$

$$\frac{1}{8} c_{\gamma} y_e^{i1i2} (-1 + 2 c_{2\gamma})(2 g_2^2 + 3 c_{2\gamma} (g_1^2+g_2^2))$$

$$
\begin{aligned}
&\left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu}\, \overline{y_e}^{pr}\right) \left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu}\, y_e^{pr}\right) \text{LF}_{4,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] + \\
&\frac{1}{6} c_\gamma y_e^{i1i2} \left(-2 g_2^2 + 3 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right)\right) \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu}\, \overline{y_e}^{pr}\right) \left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu}\, y_e^{pr}\right) \\
&\quad \text{LF}_{5,1,-2}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] - \frac{1}{2} s_{2\gamma} s_\gamma \frac{1}{m_\Phi^2} c_\gamma^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_e^{i1i2} \text{LF}_{1,1,0}\left[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}\right] - \\
&\frac{1}{2} c_\gamma^5 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_e^{i1i2} \text{LF}_{2,1,0}\left[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}\right] + \\
&\frac{1}{2} c_\gamma^5 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_e^{i1i2} \text{LF}_{3,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}\right] - \\
&\frac{1}{8} s_{4\gamma} s_\gamma \frac{1}{m_\Phi^2} \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \left(g_1^2 + g_2^2\right) \text{LF}_{1,1,0}\left[m_{\tilde{l}}^{p}, \tilde{\mu}\right] + \\
&\frac{1}{4} c_{2\gamma} c_\gamma g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,0}\left[m_{\tilde{l}}^{p}, \tilde{\mu}\right] - \\
&\frac{1}{8} c_{2\gamma} c_\gamma g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \text{LF}_{2,2,-1}\left[m_{\tilde{l}}^{p}, \tilde{\mu}\right] + \\
&\frac{1}{16} c_{2\gamma} c_\gamma g_1^2 y_e^{i1i2} \left(g_1^2 - g_2^2\right) \text{LF}_{2,1,0}\left[m_{\tilde{l}}^{i1}, m_1\right] + \\
&\frac{1}{16} c_{2\gamma} c_\gamma g_2^2 y_e^{i1i2} \left(3 g_1^2 + g_2^2\right) \text{LF}_{2,1,0}\left[m_{\tilde{l}}^{i1}, m_2\right] + \\
&\frac{1}{16} \frac{1}{m_\Phi^2} y_e^{i1i2} \left(s_\gamma \tilde{\mu}\, \overline{y_d}^{pr} \left(c_\gamma a_d^{pr} \left(4 c_{2\gamma} c_\gamma \left(g_1^2 (1 + 3 c_{2\gamma}) + 3 g_2^2 (1 + c_{2\gamma})\right) - s_\gamma \left(4 s_{2\gamma}\right.\right.\right.\right. \\
&\qquad\qquad \left.\left.\left(g_1^2 + 3 g_2^2\right) + 15 s_{4\gamma} \left(g_1^2 + g_2^2\right)\right)\right) - \tilde{\mu}\, y_d^{pr} \left(6 s_{4\gamma} c_\gamma^2 \left(g_1^2 + g_2^2\right) + 4 s_\gamma c_{2\gamma} c_\gamma\right. \\
&\qquad\quad \left.\left.\left(g_1^2 (1 + 3 c_{2\gamma}) + 3 g_2^2 (1 + c_{2\gamma})\right) - s_\gamma^2 \left(4 s_{2\gamma} \left(g_1^2 + 3 g_2^2\right) + 9 s_{4\gamma} \left(g_1^2 + g_2^2\right)\right)\right)\right) + \\
&\quad \overline{a_d}^{pr} \left(s_\gamma c_\gamma a_d^{pr} \left(c_\gamma \left(4 s_{2\gamma} \left(g_1^2 + 3 g_2^2\right) + 15 s_{4\gamma} \left(g_1^2 + g_2^2\right)\right) + \right.\right. \\
&\qquad\quad \left. 4 s_\gamma c_{2\gamma} \left(g_1^2 (1 + 3 c_{2\gamma}) + 3 g_2^2 (1 + c_{2\gamma})\right)\right) + \\
&\qquad \tilde{\mu}\, y_d^{pr} \left(6 s_{4\gamma} c_\gamma^3 \left(g_1^2 + g_2^2\right) - c_\gamma s_\gamma^2 \left(4 s_{2\gamma} \left(g_1^2 + 3 g_2^2\right) + 9 s_{4\gamma} \left(g_1^2 + g_2^2\right)\right) - 4 c_{2\gamma} s_\gamma^3\right. \\
&\qquad\quad \left.\left.\left.\left(g_1^2 (1 + 3 c_{2\gamma}) + 3 g_2^2 (1 + c_{2\gamma})\right)\right)\right)\right) \text{LF}_{2,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] + \frac{3}{16} \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \\
&\quad \left(s_\gamma \tilde{\mu}\, \overline{y_d}^{pr} \left(c_\gamma a_d^{pr} \left(-4 c_\gamma c_{2\gamma}^2 + 5 s_{4\gamma} s_\gamma\right) + \tilde{\mu}\, y_d^{pr} \left(2 s_{4\gamma} c_\gamma^2 + 4 s_\gamma c_\gamma c_{2\gamma}^2 - 3 s_{4\gamma} s_\gamma^2\right)\right) + \right. \\
&\qquad \left.\overline{a_d}^{pr} \left(-s_\gamma c_\gamma a_d^{pr} \left(5 s_{4\gamma} c_\gamma + 4 s_\gamma c_{2\gamma}^2\right) + \tilde{\mu}\, y_d^{pr} \left(-2 s_{4\gamma} c_\gamma^3 + 3 s_{4\gamma} c_\gamma s_\gamma^2 + 4 s_\gamma^3 c_{2\gamma}^2\right)\right)\right) \\
&\quad \text{LF}_{3,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] + \frac{1}{4} c_{2\gamma} c_\gamma y_e^{i1i2} \left(g_1^2 (1 + 3 c_{2\gamma}) + 3 g_2^2 (1 + c_{2\gamma})\right) \\
&\quad \left(c_\gamma \overline{a_d}^{pr} - s_\gamma \tilde{\mu}\, \overline{y_d}^{pr}\right) \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu}\, y_d^{pr}\right) \text{LF}_{3,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] - \\
&\frac{1}{4} c_{2\gamma} c_\gamma g_1^2 y_e^{i1i2} \left(c_\gamma \overline{a_d}^{pr} - s_\gamma \tilde{\mu}\, \overline{y_d}^{pr}\right) \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu}\, y_d^{pr}\right) \text{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] - \\
&\frac{3}{8} c_\gamma y_e^{i1i2} \left(-2 g_2^2 + c_{2\gamma} (1 + 6 c_{2\gamma}) \left(g_1^2 + g_2^2\right)\right) \left(c_\gamma \overline{a_d}^{pr} - s_\gamma \tilde{\mu}\, \overline{y_d}^{pr}\right) \\
&\quad \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu}\, y_d^{pr}\right) \text{LF}_{4,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] + \frac{1}{2} c_\gamma y_e^{i1i2} \left(-2 g_2^2 + 3 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right)\right) \\
&\quad \left(c_\gamma \overline{a_d}^{pr} - s_\gamma \tilde{\mu}\, \overline{y_d}^{pr}\right) \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu}\, y_d^{pr}\right) \text{LF}_{5,1,-2}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] + \\
&\frac{3}{2} s_{2\gamma} s_\gamma \frac{1}{m_\Phi^2} y_e^{i1i2} \left(-c_\gamma^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} + s_\gamma^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr}\right) \text{LF}_{1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] - \\
&\frac{3}{2} c_\gamma y_e^{i1i2} \left(c_\gamma^2 \overline{y_d}^{pr} y_d^{sr} \left(c_\gamma^2 \overline{y_d}^{st} y_d^{pt} - s_\gamma^2 \overline{y_u}^{st} y_u^{pt}\right) + s_\gamma^4 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr}\right) \\
&\quad \text{LF}_{2,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] + \frac{3}{2} c_\gamma y_e^{i1i2} \\
&\quad \left(c_\gamma^2 \overline{y_d}^{pr} y_d^{sr} \left(c_\gamma^2 \overline{y_d}^{st} y_d^{pt} - s_\gamma^2 \overline{y_u}^{st} y_u^{pt}\right) + s_\gamma^4 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr}\right) \text{LF}_{3,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] + \\
&\frac{3}{8} \frac{1}{m_\Phi^4} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \left(s_\gamma \tilde{\mu}\, \overline{y_u}^{pr} \left(-2 a_u^{pr} \left(c_\gamma^2 \left(c_{2\gamma}^2 + c_{4\gamma}\right) - s_{4\gamma} s_\gamma c_\gamma + s_{2\gamma}^2 s_\gamma^2\right) + \right.\right. \\
&\qquad\qquad \left.\tilde{\mu}\, y_u^{pr} \left(-s_{4\gamma} c_\gamma^2 - 2 s_\gamma c_\gamma \left(c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2\right) + s_{4\gamma} s_\gamma^2\right)\right) + \\
&\qquad \overline{a_u}^{pr} \left(2 s_\gamma c_\gamma a_u^{pr} \left(s_{4\gamma} c_\gamma + s_\gamma \left(c_{2\gamma}^2 + c_{4\gamma} - s_{2\gamma}^2\right)\right) + \right. \\
&\qquad\qquad \left.\left.\tilde{\mu}\, y_u^{pr} \left(-s_{4\gamma} c_\gamma^3 + 2 s_\gamma s_{2\gamma}^2 c_\gamma^2 + s_{4\gamma} c_\gamma s_\gamma^2 + 2 s_\gamma^3 \left(c_{2\gamma}^2 + c_{4\gamma}\right)\right)\right)\right) \\
&\quad \text{LF}_{1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \frac{1}{4} s_\gamma \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 - 3 g_2^2\right) \left(s_\gamma \overline{a_u}^{pr} - \tilde{\mu}\, c_\gamma \overline{y_u}^{pr}\right) \\
&\quad \left(a_u^{pr} (c_{2\gamma} c_\gamma - s_{2\gamma} s_\gamma) + \tilde{\mu}\, y_u^{pr} (s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma})\right) \text{LF}_{2,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \\
&\frac{1}{8} c_\gamma y_e^{i1i2} \left(c_{2\gamma} g_1^2 + g_2^2 (4 - 3 c_{2\gamma})\right) \left(-s_\gamma \overline{a_u}^{pr} + \tilde{\mu}\, c_\gamma \overline{y_u}^{pr}\right) \\
&\quad \left(-s_\gamma a_u^{pr} + \tilde{\mu}\, c_\gamma y_u^{pr}\right) \text{LF}_{2,2,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \\
&\frac{3}{2} c_{2\gamma} c_\gamma g_2^2 y_e^{i1i2} \left(-s_\gamma \overline{a_u}^{pr} + \tilde{\mu}\, c_\gamma \overline{y_u}^{pr}\right) \left(-s_\gamma a_u^{pr} + \tilde{\mu}\, c_\gamma y_u^{pr}\right) \text{LF}_{3,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \\
&\frac{1}{4} c_\gamma g_2^2 y_e^{i1i2} (-1 + 3 c_{2\gamma}) \left(-s_\gamma \overline{a_u}^{pr} + \tilde{\mu}\, c_\gamma \overline{y_u}^{pr}\right) \left(-s_\gamma a_u^{pr} + \tilde{\mu}\, c_\gamma y_u^{pr}\right) \text{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \\
&\frac{3}{2} c_{2\gamma} c_\gamma g_2^2 y_e^{i1i2} \left(-s_\gamma \overline{a_u}^{pr} + \tilde{\mu}\, c_\gamma \overline{y_u}^{pr}\right) \left(-s_\gamma a_u^{pr} + \tilde{\mu}\, c_\gamma y_u^{pr}\right) \text{LF}_{4,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \\
&\frac{3}{2} s_\gamma^2 c_\gamma^3 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \text{LF}_{2,1,0}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}\right] - \\
&\frac{3}{2} s_\gamma^2 c_\gamma^3 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \text{LF}_{3,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}\right] + \frac{1}{16} \frac{1}{m_\Phi^2} y_e^{i1i2} \left(s_\gamma \tilde{\mu}\, c_\gamma \overline{y_u}^{pr}\right. \\
&\qquad \left(a_u^{pr} \left(4 c_{2\gamma} c_\gamma \left(g_1^2 (-4 + 3 c_{2\gamma}) + 3 c_{2\gamma} g_2^2\right) + s_\gamma \left(16 s_{2\gamma} g_1^2 - 15 s_{4\gamma} \left(g_1^2 + g_2^2\right)\right)\right) + \right. \\
&\qquad\quad \left.\left.\tilde{\mu}\, y_u^{pr} \left(c_\gamma \left(-16 s_{2\gamma} g_1^2 + 15 s_{4\gamma} \left(g_1^2 + g_2^2\right)\right) + 4 s_\gamma c_{2\gamma} \left(g_1^2 (-4 + 3 c_{2\gamma}) + 3 c_{2\gamma} g_2^2\right)\right)\right) + \\
&\qquad \overline{a_u}^{pr} \left(s_\gamma a_u^{pr} \left(-6 s_{4\gamma} c_\gamma^2 \left(g_1^2 + g_2^2\right) - 4 s_\gamma c_{2\gamma} c_\gamma \left(g_1^2 (-4 + 3 c_{2\gamma}) + 3 c_{2\gamma} g_2^2\right) + \right.\right. \\
&\qquad\qquad \left.s_\gamma^2 \left(-16 s_{2\gamma} g_1^2 + 9 s_{4\gamma} \left(g_1^2 + g_2^2\right)\right)\right) + \tilde{\mu}\, y_u^{pr} \left(6 s_{4\gamma} c_\gamma^3 \left(g_1^2 + g_2^2\right) + \right. \\
&\qquad\qquad \left.\left.\left.c_\gamma s_\gamma^2 \left(16 s_{2\gamma} g_1^2 - 9 s_{4\gamma} \left(g_1^2 + g_2^2\right)\right) - 4 c_{2\gamma} s_\gamma^3 \left(g_1^2 (-4 + 3 c_{2\gamma}) + 3 c_{2\gamma} g_2^2\right)\right)\right)\right) \\
&\quad \text{LF}_{2,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] + \frac{3}{16} \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \left(-s_\gamma \tilde{\mu}\, c_\gamma \overline{y_u}^{pr} \left(a_u^{pr} \left(4 c_\gamma c_{2\gamma}^2 - 5 s_{4\gamma} s_\gamma\right) + \right.\right. \\
&\qquad \left.\tilde{\mu}\, y_u^{pr} \left(5 s_{4\gamma} c_\gamma + 4 s_\gamma c_{2\gamma}^2\right)\right) + \overline{a_u}^{pr} \left(s_\gamma a_u^{pr} \left(2 s_{4\gamma} c_\gamma^2 + 4 s_\gamma c_\gamma c_{2\gamma}^2 - 3 s_{4\gamma} s_\gamma^2\right) + \right. \\
&\qquad \left.\left.\tilde{\mu}\, y_u^{pr} \left(-2 s_{4\gamma} c_\gamma^3 + 3 s_{4\gamma} c_\gamma s_\gamma^2 + 4 s_\gamma^3 c_{2\gamma}^2\right)\right)\right) \text{LF}_{3,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] + \\
&\frac{1}{4} c_\gamma y_e^{i1i2} \left(c_{2\gamma} g_1^2 (-4 + 3 c_{2\gamma}) + g_2^2 \left(-2 + 3 c_{2\gamma}^2\right)\right) \left(-s_\gamma \overline{a_u}^{pr} + \tilde{\mu}\, c_\gamma \overline{y_u}^{pr}\right) \\
&\quad \left(-s_\gamma a_u^{pr} + \tilde{\mu}\, c_\gamma y_u^{pr}\right) \text{LF}_{3,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] - \\
&\frac{1}{8} c_\gamma y_e^{i1i2} \left(c_{2\gamma} g_1^2 + g_2^2 (4 - 3 c_{2\gamma})\right) \left(-s_\gamma \overline{a_u}^{pr} + \tilde{\mu}\, c_\gamma \overline{y_u}^{pr}\right) \\
&\quad \left(-s_\gamma a_u^{pr} + \tilde{\mu}\, c_\gamma y_u^{pr}\right) \text{LF}_{3,2,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] - \\
&\frac{3}{4} c_\gamma y_e^{i1i2} \left(c_{2\gamma} g_1^2 (-2 + 3 c_{2\gamma}) + g_2^2 \left(-2 + 3 c_{2\gamma}^2\right)\right) \left(-s_\gamma \overline{a_u}^{pr} + \tilde{\mu}\, c_\gamma \overline{y_u}^{pr}\right) \\
&\quad \left(-s_\gamma a_u^{pr} + \tilde{\mu}\, c_\gamma y_u^{pr}\right) \text{LF}_{4,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] + \\
&\frac{1}{2} c_\gamma y_e^{i1i2} \left(-2 g_2^2 + 3 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right)\right) \left(s_\gamma \overline{a_u}^{pr} - \tilde{\mu}\, c_\gamma \overline{y_u}^{pr}\right) \\
&\quad \left(s_\gamma a_u^{pr} - \tilde{\mu}\, c_\gamma y_u^{pr}\right) \text{LF}_{5,1,-2}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] + \\
&\frac{3}{2} s_{2\gamma} \frac{1}{m_\Phi^2} s_\gamma^3 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \text{LF}_{1,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}\right] - \\
&\frac{9}{32} s_{4\gamma} s_\gamma g_1^2 \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \text{LF}_{2,1,-1}[\tilde{\mu}, m_1] + \\
&\frac{1}{8} m_1 \tilde{\mu}\, g_1^2 \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \\
&\quad \left(s_{4\gamma} c_\gamma^3 + 2 s_\gamma c_{2\gamma}^2 c_\gamma^2 - 4 s_{4\gamma} c_\gamma s_\gamma^2 - 2 s_\gamma^3 c_{2\gamma}^2\right) \text{LF}_{2,1,0}[\tilde{\mu}, m_1] + \\
&\frac{3}{16} s_{4\gamma} s_\gamma g_1^2 \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \text{LF}_{3,1,-2}[\tilde{\mu}, m_1] + \frac{1}{8} g_1^2 \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \\
&\quad \left(-m_1 s_{4\gamma} \tilde{\mu}\, c_\gamma \left(c_\gamma^2 - 4 s_\gamma^2\right) + 2 c_{2\gamma}^2 \left(-2 c_\gamma m_\Phi^2 + m_1 s_\gamma \tilde{\mu} \left(-c_\gamma^2 + s_\gamma^2\right)\right)\right) \text{LF}_{3,1,-1}[\tilde{\mu}, m_1] - \\
&\frac{1}{2} m_1 s_\gamma \tilde{\mu}\, g_1^2 c_{2\gamma}^2 c_\gamma^2 y_e^{i1i2} \left(g_1^2 + g_2^2\right) \text{LF}_{3,1,0}[\tilde{\mu}, m_1] + \\
&\frac{3}{8} c_\gamma g_1^4 y_e^{i1i2} \text{LF}_{3,2,-2}[\tilde{\mu}, m_1] + \\
&\frac{1}{8} c_\gamma g_1^4 y_e^{i1i2} \left(3 m_1^2 + 20 m_1 s_\gamma \tilde{\mu}\, c_\gamma + 8 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) \text{LF}_{3,2,-1}[\tilde{\mu}, m_1] + \\
&g_1^4 m_1^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 y_e^{i1i2} \text{LF}_{3,2,0}[\tilde{\mu}, m_1] + \\
&\frac{1}{12} c_\gamma g_1^2 y_e^{i1i2} \left(-5 g_2^2 + 12 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right)\right) \text{LF}_{4,1,-2}[\tilde{\mu}, m_1] + \\
&\frac{1}{2} m_1 s_\gamma \tilde{\mu}\, g_1^2 c_\gamma^2 y_e^{i1i2} \left(-g_2^2 + 3 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right)\right) \text{LF}_{4,1,-1}[\tilde{\mu}, m_1] - \\
&\frac{1}{4} c_\gamma g_1^4 y_e^{i1i2} \text{LF}_{4,2,-3}[\tilde{\mu}, m_1] - \\
&\frac{1}{4} c_\gamma g_1^4 y_e^{i1i2} \left(m_1^2 + 8 m_1 s_\gamma \tilde{\mu}\, c_\gamma + 4 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) \text{LF}_{4,2,-2}[\tilde{\mu}, m_1] - \\
&g_1^4 m_1^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 y_e^{i1i2} \text{LF}_{4,2,-1}[\tilde{\mu}, m_1] - \\
&\frac{1}{6} c_\gamma g_1^2 y_e^{i1i2} \left(-2 g_2^2 + 3 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right)\right) \text{LF}_{5,1,-3}[\tilde{\mu}, m_1] + \\
&\frac{1}{3} m_1 s_\gamma \tilde{\mu}\, g_1^2 c_\gamma^2 y_e^{i1i2} \left(2 g_2^2 - 3 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right)\right) \text{LF}_{5,1,-2}[\tilde{\mu}, m_1] - \\
&\frac{27}{32} s_{4\gamma} s_\gamma g_2^2 \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \text{LF}_{2,1,-1}[\tilde{\mu}, m_2] + \\
&\frac{3}{8} m_2 \tilde{\mu}\, g_2^2 \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \left(s_{4\gamma} c_\gamma^3 + 2 s_\gamma c_{2\gamma}^2 c_\gamma^2 - 4 s_{4\gamma} c_\gamma s_\gamma^2 - 2 s_\gamma^3 c_{2\gamma}^2\right) \\
&\quad \text{LF}_{2,1,0}[\tilde{\mu}, m_2] + \frac{9}{16} s_{4\gamma} s_\gamma g_2^2 \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \text{LF}_{3,1,-2}[\tilde{\mu}, m_2] + \\
&\frac{1}{24} g_2^2 \frac{1}{m_\Phi^2} y_e^{i1i2} \left(-4 c_\gamma m_\Phi^2 \left(-8 g_2^2 + 9 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right)\right) - \right. \\
&\qquad \left. 9 m_2 \tilde{\mu} \left(g_1^2 + g_2^2\right) \left(s_{4\gamma} c_\gamma^3 + 2 s_\gamma c_{2\gamma}^2 c_\gamma^2 - 4 s_{4\gamma} c_\gamma s_\gamma^2 - 2 s_\gamma^3 c_{2\gamma}^2\right)\right) \text{LF}_{3,1,-1}[\tilde{\mu}, m_2] + \\
&\frac{1}{6} m_2 s_\gamma \tilde{\mu}\, g_2^2 c_\gamma^2 y_e^{i1i2} \left(8 g_2^2 - 9 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right)\right) \text{LF}_{3,1,0}[\tilde{\mu}, m_2] + \\
&\frac{1}{24} c_\gamma g_2^4 y_e^{i1i2} \left(69 c_\gamma^4 + 16 s_\gamma^2 + 69 s_\gamma^4 + c_\gamma^2 \left(16 - 6 s_\gamma^2\right)\right) \text{LF}_{3,2,-2}[\tilde{\mu}, m_2] + \frac{1}{24} c_\gamma g_2^4 \\
&\quad y_e^{i1i2} \left(3 m_2^2 \left(3 c_\gamma^4 + 38 s_\gamma^2 c_\gamma^2 + 3 s_\gamma^4\right) + 4 m_2 s_\gamma \tilde{\mu}\, c_\gamma \left(8 + 51 c_\gamma^2 + 51 s_\gamma^2\right) + 72 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) \\
&\quad \text{LF}_{3,2,-1}[\tilde{\mu}, m_2] + 3 g_2^4 m_2^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 y_e^{i1i2} \text{LF}_{3,2,0}[\tilde{\mu}, m_2] + \\
&\frac{3}{4} c_\gamma g_2^2 y_e^{i1i2} \left(-3 g_2^2 + 4 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right)\right) \text{LF}_{4,1,-2}[\tilde{\mu}, m_2] + \\
&\frac{1}{2} m_2 s_\gamma \tilde{\mu}\, g_2^2 c_\gamma^2 y_e^{i1i2} \left(-7 g_2^2 + 9 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right)\right) \text{LF}_{4,1,-1}[\tilde{\mu}, m_2] - \\
&\frac{1}{4} c_\gamma g_2^4 y_e^{i1i2} \left(7 c_\gamma^4 + 2 s_\gamma^2 c_\gamma^2 + 7 s_\gamma^4\right) \text{LF}_{4,2,-3}[\tilde{\mu}, m_2] - \\
&\frac{1}{4} c_\gamma g_2^4 y_e^{i1i2} \left(m_2^2 \left(c_\gamma^4 + 14 s_\gamma^2 c_\gamma^2 + s_\gamma^4\right) + 32 m_2 s_\gamma \tilde{\mu}\, c_\gamma + 16 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) \text{LF}_{4,2,-2}[\tilde{\mu}, m_2] - \\
&4 g_2^4 m_2^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 y_e^{i1i2} \text{LF}_{4,2,-1}[\tilde{\mu}, m_2] - \\
&\frac{1}{2} c_\gamma g_2^2 y_e^{i1i2} \left(-2 g_2^2 + 3 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right)\right) \text{LF}_{5,1,-3}[\tilde{\mu}, m_2] + \\
&m_2 s_\gamma \tilde{\mu}\, g_2^2 c_\gamma^2 y_e^{i1i2} \left(2 g_2^2 - 3 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right)\right) \text{LF}_{5,1,-2}[\tilde{\mu}, m_2] + \\
&\frac{1}{32} s_{4\gamma} s_\gamma \frac{1}{m_\Phi^2} \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \left(g_1^2 + g_2^2\right) \text{LF}_{2,1,-1}\left[\tilde{\mu}, m_{\tilde{e}}^{r}\right] - \\
&\frac{1}{2} c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \text{LF}_{2,1,0}\left[\tilde{\mu}, m_{\tilde{e}}^{r}\right] + \\
&c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \text{LF}_{3,1,-1}\left[\tilde{\mu}, m_{\tilde{e}}^{r}\right] - \\
&\frac{1}{2} c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \text{LF}_{4,1,-2}\left[\tilde{\mu}, m_{\tilde{e}}^{r}\right] + \\
&\frac{1}{16} s_{4\gamma} s_\gamma \frac{1}{m_\Phi^2} \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \left(g_1^2 + g_2^2\right) \text{LF}_{2,1,-1}\left[\tilde{\mu}, m_{\tilde{l}}^{p}\right] - \\
&c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \text{LF}_{2,1,0}\left[\tilde{\mu}, m_{\tilde{l}}^{s}\right] + \\
&2 c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \text{LF}_{3,1,-1}\left[\tilde{\mu}, m_{\tilde{l}}^{s}\right] - \\
&c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \text{LF}_{4,1,-2}\left[\tilde{\mu}, m_{\tilde{l}}^{s}\right] + \\
&\frac{1}{2} m_1 g_1^2 c_\gamma^2 y_e^{pi2} y_e^{i1r} \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu}\, \overline{y_e}^{pr}\right) \text{LF}_{2,1,1,0}\left[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right] + \\
&\frac{1}{2} m_1 g_1^2 c_\gamma^2 y_e^{pi2} y_e^{i1r} \left(-c_\gamma \overline{a_e}^{pr} + s_\gamma \tilde{\mu}\, \overline{y_e}^{pr}\right) \text{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right] + \\
&\frac{1}{4} m_1 g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} \left(c_\gamma a_e^{i1r} - s_\gamma \tilde{\mu}\, y_e^{i1r}\right) \text{LF}_{2,1,1,0}\left[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}\right] + \\
&\frac{1}{8} m_1 g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} \left(c_\gamma a_e^{i1r} - s_\gamma \tilde{\mu}\, y_e^{i1r}\right) \text{LF}_{2,2,1,-1}\left[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}\right] + \\
&\frac{1}{4} m_1 g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} \left(-c_\gamma a_e^{i1r} + s_\gamma \tilde{\mu}\, y_e^{i1r}\right) \text{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}\right] - \\
&g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \text{LF}_{1,1,1,0}\left[m_1, m_{\tilde{e}}^{r}, \tilde{\mu}\right] + \\
&2 g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,1,-1}\left[m_1, m_{\tilde{e}}^{r}, \tilde{\mu}\right] + \\
&m_1 s_\gamma \tilde{\mu}\, g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,1,0}\left[m_1, m_{\tilde{e}}^{r}, \tilde{\mu}\right] - \\
&g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \text{LF}_{3,1,1,-2}\left[m_1, m_{\tilde{e}}^{r}, \tilde{\mu}\right] - \\
&m_1 s_\gamma \tilde{\mu}\, g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \text{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{e}}^{r}, \tilde{\mu}\right] +
\end{aligned}
$$

$$g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}] -$$
$$\frac{1}{2} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}] +$$
$$\frac{1}{4} m_1 g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{i1r} \left(c_\gamma a_e^{pi2} - s_\gamma \tilde{\mu} y_e^{pi2}\right) \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{p}] +$$
$$\frac{1}{8} m_1 g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{i1r} \left(-c_\gamma a_e^{pi2} + s_\gamma \tilde{\mu} y_e^{pi2}\right) \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{p}] +$$
$$\frac{1}{4} m_1 g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{i1r} \left(-c_\gamma a_e^{pi2} + s_\gamma \tilde{\mu} y_e^{pi2}\right) \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{p}] +$$
$$\frac{1}{8} m_1 s_{4\gamma} g_1^2 \frac{1}{m_\Phi^2} \left(g_1^2 + g_2^2\right) \left(s_\gamma a_e^{i1i2} + \tilde{\mu} c_\gamma y_e^{i1i2}\right) \mathrm{LF}_{1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}] -$$
$$\frac{1}{16} m_1 g_1^2 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right) \left(c_\gamma a_e^{i1i2} - s_\gamma \tilde{\mu} y_e^{i1i2}\right) \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}] +$$
$$\frac{1}{8} s_{4\gamma} s_\gamma g_1^2 \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \mathrm{LF}_{1,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}] -$$
$$\frac{1}{8} m_1 s_{4\gamma} \tilde{\mu} c_\gamma g_1^2 \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \mathrm{LF}_{1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}] +$$
$$\frac{3}{4} c_\gamma g_1^4 y_e^{i1i2} \mathrm{LF}_{2,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}] +$$
$$\frac{1}{2} m_1 c_\gamma g_1^4 y_e^{i1i2} (m_1 + 4 s_\gamma \tilde{\mu} c_\gamma) \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}] -$$
$$\frac{1}{2} c_\gamma g_1^4 y_e^{i1i2} \mathrm{LF}_{3,1,1,-2}[m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}] -$$
$$\frac{1}{2} m_1 c_\gamma g_1^4 y_e^{i1i2} (m_1 + 4 s_\gamma \tilde{\mu} c_\gamma) \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}] +$$
$$\frac{1}{8} m_1 g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{i1r} \left(c_\gamma a_e^{pi2} - s_\gamma \tilde{\mu} y_e^{pi2}\right) \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{i2}] +$$
$$\frac{1}{2} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{1,1,1,0}[m_1, m_{\tilde{l}}^{p}, \tilde{\mu}] -$$
$$g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_1, m_{\tilde{l}}^{p}, \tilde{\mu}] -$$
$$\frac{1}{2} m_1 s_\gamma \tilde{\mu} g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{l}}^{p}, \tilde{\mu}] +$$
$$\frac{1}{2} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}[m_1, m_{\tilde{l}}^{p}, \tilde{\mu}] +$$
$$\frac{1}{2} m_1 s_\gamma \tilde{\mu} g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{l}}^{p}, \tilde{\mu}] +$$
$$\frac{1}{8} m_1 g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} \left(-c_\gamma a_e^{i1r} + s_\gamma \tilde{\mu} y_e^{i1r}\right) \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{l}}^{i1}, m_{\tilde{e}}^{r}] -$$
$$\frac{1}{16} m_1 g_1^2 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right) \left(c_\gamma a_e^{i1i2} - s_\gamma \tilde{\mu} y_e^{i1i2}\right) \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{l}}^{i1}, m_{\tilde{e}}^{i2}] +$$
$$\frac{1}{4} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{1,1,1,0}[m_1, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] -$$
$$\frac{1}{8} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_1, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] -$$
$$\frac{1}{16} s_{4\gamma} s_\gamma g_1^2 \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \mathrm{LF}_{1,1,1,-1}[m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] +$$
$$\frac{1}{16} m_1 s_{4\gamma} \tilde{\mu} c_\gamma g_1^2 \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \mathrm{LF}_{1,1,1,0}[m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] +$$
$$\frac{3}{16} c_\gamma g_1^4 y_e^{i1i2} \mathrm{LF}_{2,1,1,-1}[m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] +$$
$$\frac{1}{8} m_1 c_\gamma g_1^4 y_e^{i1i2} (m_1 + 4 s_\gamma \tilde{\mu} c_\gamma) \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] -$$
$$\frac{1}{8} c_\gamma g_1^4 y_e^{i1i2} \mathrm{LF}_{3,1,1,-2}[m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] -$$
$$\frac{1}{8} m_1 c_\gamma g_1^4 y_e^{i1i2} (m_1 + 4 s_\gamma \tilde{\mu} c_\gamma) \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] -$$
$$\frac{7}{8} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{e}}^{r}] -$$
$$\frac{1}{8} s_\gamma \tilde{\mu} c_\gamma g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} (8 m_1 c_\gamma + s_\gamma \tilde{\mu}) \mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{e}}^{r}] +$$
$$\frac{1}{8} c_\gamma g_1^2 \left(-2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} + y_e^{i1i2} \left(-6 g_1^2 + c_{2\gamma}^2 \left(g_1^2 + g_2^2\right)\right)\right) \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{e}}^{i2}] +$$
$$\frac{1}{8} \Big(-6 c_\gamma g_1^4 m_1^2 y_e^{i1i2} + m_1 s_\gamma \tilde{\mu} g_1^2 \Big(-2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} +$$
$$y_e^{i1i2} \left(c_{2\gamma}^2 \left(g_1^2 + g_2^2\right) - 8 g_1^2 \left(4 c_\gamma^2 + s_\gamma^2\right)\right)\Big) - 8 c_\gamma g_1^4 s_\gamma^2 \tilde{\mu}^2 y_e^{i1i2}\Big)$$
$$\mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{e}}^{i2}] - c_\gamma g_1^4 m_1^2 s_\gamma^2 \tilde{\mu}^2 y_e^{i1i2} \mathrm{LF}_{2,2,1,0}[m_1, \tilde{\mu}, m_{\tilde{e}}^{i2}] +$$
$$\frac{1}{4} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{l}}^{p}] +$$
$$\frac{1}{8} s_\gamma \tilde{\mu} c_\gamma g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} (m_1 c_\gamma - s_\gamma \tilde{\mu}) \mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{l}}^{p}] +$$
$$\frac{1}{16} c_\gamma g_1^2 \left(2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} - y_e^{i1i2} \left(-3 g_1^2 + c_{2\gamma}^2 \left(g_1^2 + g_2^2\right)\right)\right) \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{l}}^{i1}] +$$
$$\frac{1}{16} \Big(3 c_\gamma g_1^4 m_1^2 y_e^{i1i2} + m_1 s_\gamma \tilde{\mu} g_1^2 \Big(2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} -$$
$$y_e^{i1i2} \left(c_{2\gamma}^2 \left(g_1^2 + g_2^2\right) - 4 g_1^2 \left(5 c_\gamma^2 + 2 s_\gamma^2\right)\right)\Big) + 8 c_\gamma g_1^4 s_\gamma^2 \tilde{\mu}^2 y_e^{i1i2}\Big)$$
$$\mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{l}}^{i1}] + \frac{1}{2} c_\gamma g_1^4 m_1^2 s_\gamma^2 \tilde{\mu}^2 y_e^{i1i2} \mathrm{LF}_{2,2,1,0}[m_1, \tilde{\mu}, m_{\tilde{l}}^{i1}] -$$
$$\frac{3}{2} g_2^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{1,1,1,0}[m_2, m_{\tilde{l}}^{p}, \tilde{\mu}] +$$
$$3 g_2^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_2, m_{\tilde{l}}^{p}, \tilde{\mu}] +$$
$$\frac{3}{2} m_2 s_\gamma \tilde{\mu} g_2^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,0}[m_2, m_{\tilde{l}}^{p}, \tilde{\mu}] -$$
$$\frac{3}{2} g_2^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}[m_2, m_{\tilde{l}}^{p}, \tilde{\mu}] -$$
$$\frac{3}{2} m_2 s_\gamma \tilde{\mu} g_2^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-1}[m_2, m_{\tilde{l}}^{p}, \tilde{\mu}] +$$
$$\frac{1}{4} g_2^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{1,1,1,0}[m_2, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] -$$
$$\frac{1}{8} g_2^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_2, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] +$$
$$\frac{3}{16} s_{4\gamma} s_\gamma g_2^2 \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \mathrm{LF}_{1,1,1,-1}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] -$$
$$\frac{3}{16} m_2 s_{4\gamma} \tilde{\mu} c_\gamma g_2^2 \frac{1}{m_\Phi^2} y_e^{i1i2} \left(g_1^2 + g_2^2\right) \mathrm{LF}_{1,1,1,0}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] +$$
$$\frac{3}{16} c_\gamma g_2^4 y_e^{i1i2} \left(5 c_\gamma^2 + s_\gamma^2\right) \mathrm{LF}_{2,1,1,-1}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] +$$
$$\frac{1}{8} m_2 c_\gamma g_2^4 y_e^{i1i2} \left(m_2 \left(c_\gamma^2 + 5 s_\gamma^2\right) + 12 s_\gamma \tilde{\mu} c_\gamma\right) \mathrm{LF}_{2,1,1,0}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] -$$
$$\frac{1}{8} c_\gamma g_2^4 y_e^{i1i2} \left(5 c_\gamma^2 + s_\gamma^2\right) \mathrm{LF}_{3,1,1,-2}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] -$$
$$\frac{1}{8} m_2 c_\gamma g_2^4 y_e^{i1i2} \left(m_2 \left(c_\gamma^2 + 5 s_\gamma^2\right) + 12 s_\gamma \tilde{\mu} c_\gamma\right) \mathrm{LF}_{3,1,1,-1}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] +$$
$$2 c_\gamma g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_1] +$$
$$\frac{1}{2} c_\gamma g_1^2 g_2^2 y_e^{i1i2} \left(3 m_1 m_2 + 2 s_\gamma \tilde{\mu} c_\gamma (7 m_1 + 5 m_2) + 12 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] +$$
$$4 m_1 m_2 g_1^2 g_2^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 y_e^{i1i2} \mathrm{LF}_{2,2,1,0}[m_2, \tilde{\mu}, m_1] -$$
$$c_\gamma g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{3,2,1,-3}[m_2, \tilde{\mu}, m_1] -$$
$$c_\gamma g_1^2 g_2^2 y_e^{i1i2} \left(m_1 m_2 + 4 s_\gamma \tilde{\mu} c_\gamma (m_1 + m_2) + 4 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) \mathrm{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] -$$
$$4 m_1 m_2 g_1^2 g_2^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 y_e^{i1i2} \mathrm{LF}_{3,2,1,-1}[m_2, \tilde{\mu}, m_1] -$$
$$\frac{1}{8} g_2^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{e}}^{r}] -$$
$$\frac{1}{8} s_\gamma \tilde{\mu} c_\gamma g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} (2 m_2 c_\gamma + s_\gamma \tilde{\mu}) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{e}}^{r}] -$$
$$\frac{3}{2} g_2^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{l}}^{p}] -$$
$$\frac{3}{8} s_\gamma \tilde{\mu} c_\gamma g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} (5 m_2 c_\gamma + s_\gamma \tilde{\mu}) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{l}}^{p}] +$$
$$\frac{1}{16} c_\gamma g_2^2 \left(-6 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} + y_e^{i1i2} \left(3 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right) - 5 g_2^2 \left(5 c_\gamma^2 + s_\gamma^2\right)\right)\right)$$
$$\mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}] + \frac{1}{16} \Big(-5 c_\gamma g_2^4 m_2^2 y_e^{i1i2} \left(c_\gamma^2 + 5 s_\gamma^2\right) +$$
$$3 m_2 s_\gamma \tilde{\mu} g_2^2 \left(-2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} + y_e^{i1i2} \left(c_{2\gamma}^2 \left(g_1^2 + g_2^2\right) - 4 g_2^2 \left(7 c_\gamma^2 + 2 s_\gamma^2\right)\right)\right) -$$
$$24 c_\gamma g_2^4 s_\gamma^2 \tilde{\mu}^2 y_e^{i1i2}\Big) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}] -$$
$$\frac{3}{2} c_\gamma g_2^4 m_2^2 s_\gamma^2 \tilde{\mu}^2 y_e^{i1i2} \mathrm{LF}_{2,2,1,0}[m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}] + \frac{3}{2} s_\gamma c_\gamma \frac{1}{m_\Phi^2} \overline{y_d}^{st} y_d^{sr} y_e^{i1i2}$$
$$\left(\tilde{\mu} \overline{y_d}^{pr} a_d^{pt} \left(-2 c_\gamma^3 + s_{2\gamma} s_\gamma\right) - \overline{a_d}^{pr} (s_{2\gamma} + 2 s_\gamma c_\gamma) \left(c_\gamma a_d^{pt} - s_\gamma \tilde{\mu} y_d^{pt}\right)\right)$$
$$\mathrm{LF}_{1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] -$$
$$\frac{3}{2} \frac{1}{m_\Phi^2} s_\gamma^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_e^{i1i2} \left(-2 c_\gamma^3 + s_{2\gamma} s_\gamma\right) \mathrm{LF}_{1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}] +$$
$$\frac{3}{4} c_\gamma^4 \overline{y_d}^{st} y_d^{sr} y_e^{i1i2} \left(s_\gamma \tilde{\mu} \overline{y_d}^{pr} a_d^{pt} + \overline{a_d}^{pr} \left(-c_\gamma a_d^{pt} + s_\gamma \tilde{\mu} y_d^{pt}\right)\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}] +$$
$$\frac{3}{2} s_\gamma \frac{1}{m_\Phi^2} \overline{y_d}^{st} y_d^{pt} y_e^{i1i2}$$
$$\left(s_\gamma \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} \left(2 c_\gamma^3 - s_{2\gamma} s_\gamma\right) - c_\gamma \overline{a_d}^{pr} (s_{2\gamma} + 2 s_\gamma c_\gamma) \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right)\right)$$
$$\mathrm{LF}_{1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] + \frac{3}{4} s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_e^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{t}] +$$
$$\frac{3}{4} c_\gamma^4 \overline{y_d}^{st} y_d^{sr} y_e^{i1i2} \left(-s_\gamma \tilde{\mu} \overline{y_d}^{pr} a_d^{pt} + \overline{a_d}^{pr} \left(c_\gamma a_d^{pt} - s_\gamma \tilde{\mu} y_d^{pt}\right)\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] +$$
$$\frac{3}{2} s_\gamma \tilde{\mu} c_\gamma \frac{1}{m_\Phi^2} \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{sr} y_e^{i1i2} a_d^{pt} \left(-2 c_\gamma^3 + s_{2\gamma} s_\gamma\right) \mathrm{LF}_{1,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] -$$
$$\frac{3}{4} s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_e^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}] +$$
$$\frac{1}{2} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_{\tilde{e}}^{r}, m_1, m_{\tilde{e}}^{i2}] + \frac{1}{2} s_\gamma c_\gamma \frac{1}{m_\Phi^2} \overline{y_e}^{st} y_e^{sr} y_e^{i1i2}$$
$$\left(\tilde{\mu} \overline{y_e}^{pr} a_e^{pt} \left(-2 c_\gamma^3 + s_{2\gamma} s_\gamma\right) - \overline{a_e}^{pr} (s_{2\gamma} + 2 s_\gamma c_\gamma) \left(c_\gamma a_e^{pt} - s_\gamma \tilde{\mu} y_e^{pt}\right)\right)$$
$$\mathrm{LF}_{1,1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}] -$$
$$\frac{1}{2} \frac{1}{m_\Phi^2} s_\gamma^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_e^{i1i2} \left(-2 c_\gamma^3 + s_{2\gamma} s_\gamma\right) \mathrm{LF}_{1,1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}] +$$
$$\frac{1}{2} c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \mathrm{LF}_{1,1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, \tilde{\mu}] +$$
$$\frac{1}{4} c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \mathrm{LF}_{2,1,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, \tilde{\mu}] +$$
$$\frac{1}{4} c_\gamma^4 \overline{y_e}^{st} y_e^{sr} y_e^{i1i2} \left(s_\gamma \tilde{\mu} \overline{y_e}^{pr} a_e^{pt} + \overline{a_e}^{pr} \left(-c_\gamma a_e^{pt} + s_\gamma \tilde{\mu} y_e^{pt}\right)\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{t}] +$$
$$\frac{1}{2} s_\gamma \frac{1}{m_\Phi^2} \overline{y_e}^{st} y_e^{pt} y_e^{i1i2}$$
$$\left(s_\gamma \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{sr} \left(2 c_\gamma^3 - s_{2\gamma} s_\gamma\right) - c_\gamma \overline{a_e}^{pr} (s_{2\gamma} + 2 s_\gamma c_\gamma) \left(c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}\right)\right)$$
$$\mathrm{LF}_{1,1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}] + \frac{1}{4} s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_e^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{t}] -$$
$$\frac{1}{4} c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \mathrm{LF}_{2,1,1,-1}[m_{\tilde{e}}^{t}, m_{\tilde{e}}^{r}, \tilde{\mu}] +$$
$$\frac{1}{4} c_\gamma^4 \overline{y_e}^{st} y_e^{sr} y_e^{i1i2} \left(-s_\gamma \tilde{\mu} \overline{y_e}^{pr} a_e^{pt} + \overline{a_e}^{pr} \left(c_\gamma a_e^{pt} - s_\gamma \tilde{\mu} y_e^{pt}\right)\right) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] +$$
$$\frac{1}{2} s_\gamma \tilde{\mu} c_\gamma \frac{1}{m_\Phi^2} \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{sr} y_e^{i1i2} a_e^{pt} \left(-2 c_\gamma^3 + s_{2\gamma} s_\gamma\right) \mathrm{LF}_{1,1,1,0}[m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}] -$$
$$\frac{1}{4} s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_e^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}] -$$
$$\frac{1}{2} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_{\tilde{e}}^{i2}, m_1, m_{\tilde{e}}^{r}] +$$
$$\frac{1}{8} m_1 c_{2\gamma} g_1^2 \left(g_1^2 (4 + c_{2\gamma}) + c_{2\gamma} g_2^2\right) \left(c_\gamma a_e^{i1i2} - s_\gamma \tilde{\mu} y_e^{i1i2}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde{e}}^{i2}, m_1, m_{\tilde{l}}^{i1}] -$$
$$\frac{1}{8} m_1 g_1^2 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right) \left(c_\gamma a_e^{i1i2} - s_\gamma \tilde{\mu} y_e^{i1i2}\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{e}}^{i2}, m_1, m_{\tilde{l}}^{i1}] +$$
$$\frac{1}{2} c_{2\gamma} c_\gamma g_1^4 y_e^{i1i2} \mathrm{LF}_{2,1,1,-1}[m_{\tilde{e}}^{i2}, m_1, \tilde{\mu}] +$$
$$\frac{1}{2} m_1 s_\gamma \tilde{\mu} c_{2\gamma} g_1^4 y_e^{i1i2} \mathrm{LF}_{2,1,1,0}[m_{\tilde{e}}^{i2}, m_1, \tilde{\mu}] +$$
$$\frac{1}{8} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_{\tilde{l}}^{p}, m_1, m_{\tilde{l}}^{i1}] +$$
$$\frac{1}{8} g_2^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_{\tilde{l}}^{p}, m_2, m_{\tilde{l}}^{i1}] +$$
$$\frac{1}{2} c_\gamma^4 \overline{y_e}^{st} y_e^{sr} y_e^{i1i2} \left(s_\gamma \tilde{\mu} \overline{y_e}^{pr} a_e^{pt} + \overline{a_e}^{pr} \left(-c_\gamma a_e^{pt} + s_\gamma \tilde{\mu} y_e^{pt}\right)\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}] +$$
$$\frac{1}{2} c_\gamma^4 \overline{y_e}^{st} y_e^{sr} y_e^{i1i2} \left(-s_\gamma \tilde{\mu} \overline{y_e}^{pr} a_e^{pt} + \overline{a_e}^{pr} \left(c_\gamma a_e^{pt} - s_\gamma \tilde{\mu} y_e^{pt}\right)\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}] -$$
$$c_\gamma^3 \overline{y_e}^{st} y_e^{pt} y_e^{i1i2} \left(s_\gamma^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{sr} + c_\gamma \overline{a_e}^{pr} \left(c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}\right)\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}] +$$
$$c_\gamma^3 \overline{y_e}^{st} y_e^{pt} y_e^{i1i2} \left(s_\gamma^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{sr} + c_\gamma \overline{a_e}^{pr} \left(c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}\right)\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}] +$$
$$\frac{1}{2} c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{si2} y_e^{i1r} \mathrm{LF}_{1,1,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, \tilde{\mu}] +$$
$$\frac{1}{4} c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} \left(y_e^{pt} y_e^{si2} y_e^{i1r} - y_e^{pi2} y_e^{sr} y_e^{i1t}\right) \mathrm{LF}_{2,1,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, \tilde{\mu}] -$$
$$\frac{1}{2} s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_e^{i1i2} \mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}] +$$

$$\frac{1}{2} s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_e^{i1i2} \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}\right] +$$

$$s_\gamma \tilde{\mu} c_\gamma^4 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{sr} y_e^{i1i2} a_e^{pt} \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}\right] -$$

$$s_\gamma \tilde{\mu} c_\gamma^4 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{sr} y_e^{i1i2} a_e^{pt} \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}\right] +$$

$$\frac{1}{8} m_1 c_{2\gamma} g_1^2 \left(g_1^2 \left(-2 + c_{2\gamma}\right) + g_2^2 \left(2 + c_{2\gamma}\right)\right) \left(c_\gamma a_e^{i1i2} - s_\gamma \tilde{\mu} y_e^{i1i2}\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{i2}\right] -$$

$$\frac{1}{8} m_1 g_1^2 c_{2\gamma}^2 \left(g_1^2 + g_2^2\right) \left(c_\gamma a_e^{i1i2} - s_\gamma \tilde{\mu} y_e^{i1i2}\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{i2}\right] -$$

$$\frac{1}{8} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}\left[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{l}}^{p}\right] +$$

$$\frac{1}{8} c_{2\gamma} c_\gamma g_1^2 y_e^{i1i2} \left(g_1^2 - g_2^2\right) \mathrm{LF}_{2,1,1,-1}\left[m_{\tilde{l}}^{i1}, m_1, \tilde{\mu}\right] +$$

$$\frac{1}{8} m_1 s_\gamma \tilde{\mu} c_{2\gamma} g_1^2 y_e^{i1i2} \left(g_1^2 - g_2^2\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{l}}^{i1}, m_1, \tilde{\mu}\right] -$$

$$\frac{1}{8} g_2^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}\left[m_{\tilde{l}}^{i1}, m_2, m_{\tilde{l}}^{p}\right] -$$

$$\frac{1}{8} c_{2\gamma} c_\gamma g_2^2 y_e^{i1i2} \left(3 g_1^2 + g_2^2\right) \mathrm{LF}_{2,1,1,-1}\left[m_{\tilde{l}}^{i1}, m_2, \tilde{\mu}\right] -$$

$$\frac{1}{8} m_2 s_\gamma \tilde{\mu} c_{2\gamma} g_2^2 y_e^{i1i2} \left(3 g_1^2 + g_2^2\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{l}}^{i1}, m_2, \tilde{\mu}\right] +$$

$$\frac{3}{2} c_\gamma^4 \overline{y_d}^{st} y_d^{sr} y_e^{i1i2} \left(s_\gamma \tilde{\mu} \overline{y_d}^{pr} a_d^{pt} + \overline{a_d}^{pr} \left(-c_\gamma a_d^{pt} + s_\gamma \tilde{\mu} y_d^{pt}\right)\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}\right] +$$

$$\frac{3}{2} c_\gamma^4 \overline{y_d}^{st} y_d^{sr} y_e^{i1i2} \left(-s_\gamma \tilde{\mu} \overline{y_d}^{pr} a_d^{pt} + \overline{a_d}^{pr} \left(c_\gamma a_d^{pt} - s_\gamma \tilde{\mu} y_d^{pt}\right)\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}\right] -$$

$$3 c_\gamma^3 \overline{y_d}^{st} y_d^{pt} y_e^{i1i2} \left(s_\gamma^2 \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} + c_\gamma \overline{a_d}^{pr} \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right)\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}\right] +$$

$$3 c_\gamma^3 \overline{y_d}^{st} y_d^{pt} y_e^{i1i2} \left(s_\gamma^2 \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} + c_\gamma \overline{a_d}^{pr} \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right)\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}\right] +$$

$$\frac{3}{2} c_\gamma s_\gamma^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \left(c_\gamma \overline{a_d}^{pr} - s_\gamma \tilde{\mu} \overline{y_d}^{pr}\right) \left(c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}\right) \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}\right] +$$

$$\frac{3}{2} s_\gamma \frac{1}{m_\Phi^2} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt}$$

$$\left(c_\gamma \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{sr} \left(s_{2\gamma} c_\gamma - 2 s_\gamma^3\right) + s_\gamma \overline{a_u}^{pr} \left(s_\gamma a_u^{sr} \left(s_{2\gamma} + 2 s_\gamma c_\gamma\right) + \tilde{\mu} y_u^{sr} \left(-s_{2\gamma} c_\gamma + 2 s_\gamma^3\right)\right)\right)$$

$$\mathrm{LF}_{1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}\right] - \frac{3}{2} s_\gamma c_\gamma y_e^{i1i2}$$

$$\left(s_\gamma \tilde{\mu}^2 c_\gamma^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} + \overline{a_u}^{pr} \left(c_\gamma^2 \overline{y_d}^{st} y_d^{pt} - s_\gamma^2 \overline{y_u}^{st} y_u^{pt}\right) \left(-s_\gamma a_u^{sr} + \tilde{\mu} c_\gamma y_u^{sr}\right)\right)$$

$$\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}\right] + \frac{3}{2} s_\gamma c_\gamma y_e^{i1i2}$$

$$\left(s_\gamma \tilde{\mu}^2 c_\gamma^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} + \overline{a_u}^{pr} \left(c_\gamma^2 \overline{y_d}^{st} y_d^{pt} - s_\gamma^2 \overline{y_u}^{st} y_u^{pt}\right) \left(-s_\gamma a_u^{sr} + \tilde{\mu} c_\gamma y_u^{sr}\right)\right)$$

$$\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}\right] -$$

$$\frac{3}{2} \tilde{\mu} c_\gamma \frac{1}{m_\Phi^2} s_\gamma^2 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} \left(s_{2\gamma} + 2 s_\gamma c_\gamma\right) \mathrm{LF}_{1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] +$$

$$\frac{3}{2} \tilde{\mu} c_\gamma^2 y_e^{i1i2} \overline{y_u}^{st} \left(c_\gamma^2 \overline{y_d}^{pr} y_d^{sr} - s_\gamma^2 \overline{y_u}^{pr} y_u^{sr}\right) \left(-s_\gamma a_u^{pt} + \tilde{\mu} c_\gamma y_u^{pt}\right)$$

$$\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] + \frac{3}{2} \tilde{\mu} c_\gamma^2 y_e^{i1i2} \overline{y_u}^{st} \left(c_\gamma^2 \overline{y_d}^{pr} y_d^{sr} - s_\gamma^2 \overline{y_u}^{pr} y_u^{sr}\right)$$

$$\left(s_\gamma a_u^{pt} - \tilde{\mu} c_\gamma y_u^{pt}\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] + \frac{3}{4} s_\gamma c_\gamma y_e^{i1i2}$$

$$\left(2 s_\gamma \tilde{\mu}^2 c_\gamma^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} + \overline{a_u}^{pr} \left(c_\gamma^2 \overline{y_d}^{st} y_d^{pt} - 2 s_\gamma^2 \overline{y_u}^{st} y_u^{pt}\right) \left(-s_\gamma a_u^{sr} + \tilde{\mu} c_\gamma y_u^{sr}\right)\right)$$

$$\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}\right] + \frac{3}{2} \frac{1}{m_\Phi^2} s_\gamma^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{sr}$$

$$\left(-\tilde{\mu} c_\gamma \overline{y_u}^{pr} a_u^{pt} \left(s_{2\gamma} + 2 s_\gamma c_\gamma\right) + \overline{a_u}^{pr} \left(s_\gamma a_u^{pt} \left(s_{2\gamma} + 2 s_\gamma c_\gamma\right) + \tilde{\mu} y_u^{pt} \left(-s_{2\gamma} c_\gamma + 2 s_\gamma^3\right)\right)\right)$$

$$\mathrm{LF}_{1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}\right] +$$

$$\frac{3}{4} \tilde{\mu} c_\gamma^4 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} \left(s_\gamma a_u^{pt} - \tilde{\mu} c_\gamma y_u^{pt}\right) \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}\right] -$$

$$\frac{3}{2} s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_e^{i1i2} \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}\right] +$$

$$\frac{3}{2} s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_e^{i1i2} \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}\right] +$$

$$3 s_\gamma \tilde{\mu} c_\gamma^4 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{sr} y_e^{i1i2} a_d^{pt} \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}\right] -$$

$$3 s_\gamma \tilde{\mu} c_\gamma^4 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{sr} y_e^{i1i2} a_d^{pt} \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}\right] +$$

$$\frac{3}{2} s_\gamma c_\gamma y_e^{i1i2} \overline{a_u}^{pr} \left(c_\gamma^2 \overline{y_d}^{st} y_d^{pt} - s_\gamma^2 \overline{y_u}^{st} y_u^{pt}\right) \left(s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr}\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] +$$

$$\frac{3}{2} s_\gamma c_\gamma y_e^{i1i2} \overline{a_u}^{pr} \left(c_\gamma^2 \overline{y_d}^{st} y_d^{pt} - s_\gamma^2 \overline{y_u}^{st} y_u^{pt}\right)$$

$$\left(-s_\gamma a_u^{sr} + \tilde{\mu} c_\gamma y_u^{sr}\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] +$$

$$\frac{3}{2} \tilde{\mu} c_\gamma^2 y_e^{i1i2} \overline{y_u}^{st} \left(s_\gamma^3 \overline{y_u}^{pr} y_u^{sr} a_u^{pt} + c_\gamma^2 \overline{y_d}^{pr} y_d^{sr} \left(-s_\gamma a_u^{pt} + \tilde{\mu} c_\gamma y_u^{pt}\right)\right)$$

$$\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}\right] - \frac{3}{2} \tilde{\mu} c_\gamma^2 y_e^{i1i2} \overline{y_u}^{st}$$

$$\left(s_\gamma^3 \overline{y_u}^{pr} y_u^{sr} a_u^{pt} + c_\gamma^2 \overline{y_d}^{pr} y_d^{sr} \left(-s_\gamma a_u^{pt} + \tilde{\mu} c_\gamma y_u^{pt}\right)\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}\right] +$$

$$\frac{3}{4} s_\gamma c_\gamma^3 \overline{y_d}^{st} y_d^{pt} y_e^{i1i2} \overline{a_u}^{pr} \left(-s_\gamma a_u^{sr} + \tilde{\mu} c_\gamma y_u^{sr}\right) \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] +$$

$$\frac{3}{2} s_\gamma c_\gamma \frac{1}{m_\Phi^2} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \left(s_{2\gamma} c_\gamma - 2 s_\gamma^3\right) \mathrm{LF}_{1,1,1,0}\left[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}\right] +$$

$$\frac{3}{4} \tilde{\mu} c_\gamma^2 y_e^{i1i2} \overline{y_u}^{st} \left(-2 s_\gamma^3 \overline{y_u}^{pr} y_u^{sr} a_u^{pt} + c_\gamma^2 \overline{y_d}^{pr} y_d^{sr} \left(s_\gamma a_u^{pt} - \tilde{\mu} c_\gamma y_u^{pt}\right)\right)$$

$$\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}\right] - \frac{3}{2} c_\gamma s_\gamma^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt}$$

$$\left(\tilde{\mu}^2 c_\gamma^2 \overline{y_u}^{pr} y_u^{sr} + s_\gamma \overline{a_u}^{pr} \left(s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr}\right)\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] +$$

$$\frac{3}{2} c_\gamma s_\gamma^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \left(\tilde{\mu}^2 c_\gamma^2 \overline{y_u}^{pr} y_u^{sr} + s_\gamma \overline{a_u}^{pr} \left(s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr}\right)\right)$$

$$\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] - 3 s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] +$$

$$3 s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] -$$

$$\frac{3}{2} c_\gamma s_\gamma^3 y_e^{i1i2} \overline{y_u}^{st} y_u^{sr} \left(\tilde{\mu} c_\gamma \overline{y_u}^{pr} a_u^{pt} + \overline{a_u}^{pr} \left(-s_\gamma a_u^{pt} + \tilde{\mu} c_\gamma y_u^{pt}\right)\right) \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}\right] +$$

$$\frac{3}{2} s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}\right] +$$

$$\frac{3}{2} \tilde{\mu} s_\gamma^3 c_\gamma^2 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] -$$

$$\frac{3}{2} \tilde{\mu} s_\gamma^3 c_\gamma^2 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] +$$

$$3 c_\gamma s_\gamma^3 y_e^{i1i2} \overline{y_u}^{st} y_u^{sr} \left(\tilde{\mu} c_\gamma \overline{y_u}^{pr} a_u^{pt} + \overline{a_u}^{pr} \left(-s_\gamma a_u^{pt} + \tilde{\mu} c_\gamma y_u^{pt}\right)\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] -$$

$$3 c_\gamma s_\gamma^3 y_e^{i1i2} \overline{y_u}^{st} y_u^{sr} \left(\tilde{\mu} c_\gamma \overline{y_u}^{pr} a_u^{pt} + \overline{a_u}^{pr} \left(-s_\gamma a_u^{pt} + \tilde{\mu} c_\gamma y_u^{pt}\right)\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] +$$

$$\frac{1}{4} g_1^2 g_2^2 \frac{1}{m_\Phi^2} y_e^{i1i2} \left(-c_\gamma m_\Phi^2 \left(3 c_\gamma^4 + 8 s_\gamma^2 c_\gamma^2 + 3 s_\gamma^4\right) + 4 s_\gamma \tilde{\mu} \left(m_1 + m_2\right) \left(-c_\gamma^4 + s_\gamma^4\right) +\right.$$

$$\left.8 c_\gamma s_\gamma^2 \tilde{\mu}^2 \left(-c_\gamma^2 + s_\gamma^2\right)\right) \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_1, m_2\right] - m_1 s_\gamma c_\gamma g_1^2 g_2^2 \frac{1}{m_\Phi^2}$$

$$y_e^{i1i2} \left(m_2 s_\gamma m_\Phi^2 c_\gamma^2 + 2 \tilde{\mu} c_\gamma m_\Phi^2 + 2 m_2 s_\gamma \tilde{\mu}^2 \left(c_\gamma^2 - s_\gamma^2\right)\right) \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_1, m_2\right] +$$

$$\frac{1}{4} c_\gamma g_1^2 g_2^2 y_e^{i1i2} \left(7 c_\gamma^4 + 10 s_\gamma^2 c_\gamma^2 + 7 s_\gamma^4\right) \mathrm{LF}_{3,1,1,-2}\left[\tilde{\mu}, m_1, m_2\right] +$$

$$\frac{1}{4} c_\gamma g_1^2 g_2^2 y_e^{i1i2} \left(m_1 m_2 \left(3 c_\gamma^2 + s_\gamma^2\right) \left(c_\gamma^2 + 3 s_\gamma^2\right) + 4 s_\gamma \tilde{\mu} c_\gamma \left(5 m_1 + 3 m_2\right) + 16 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right)$$

$$\mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_2\right] + 2 m_1 m_2 g_1^2 g_2^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 y_e^{i1i2} \mathrm{LF}_{3,1,1,0}\left[\tilde{\mu}, m_1, m_2\right] -$$

$$c_\gamma g_1^2 g_2^2 y_e^{i1i2} \left(c_\gamma^4 + s_\gamma^2 c_\gamma^2 + s_\gamma^4\right) \mathrm{LF}_{4,1,1,-3}\left[\tilde{\mu}, m_1, m_2\right] +$$

$$\frac{1}{2} c_\gamma g_1^2 g_2^2 y_e^{i1i2} \left(-m_1 m_2 \left(c_\gamma^4 + 4 s_\gamma^2 c_\gamma^2 + s_\gamma^4\right) - 6 s_\gamma \tilde{\mu} c_\gamma \left(m_1 + m_2\right) - 6 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right)$$

$$\mathrm{LF}_{4,1,1,-2}\left[\tilde{\mu}, m_1, m_2\right] - 3 m_1 m_2 g_1^2 g_2^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 y_e^{i1i2} \mathrm{LF}_{4,1,1,-1}\left[\tilde{\mu}, m_1, m_2\right] +$$

$$\frac{13}{8} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{r}\right] +$$

$$\frac{1}{4} s_\gamma \tilde{\mu} c_\gamma g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \left(4 m_1 c_\gamma + s_\gamma \tilde{\mu}\right) \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{r}\right] -$$

$$\frac{3}{4} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{r}\right] -$$

$$\frac{1}{4} s_\gamma \tilde{\mu} c_\gamma g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \left(4 m_1 c_\gamma + s_\gamma \tilde{\mu}\right) \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{r}\right] +$$

$$\frac{3}{4} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{i2}\right] +$$

$$\frac{1}{2} m_1 s_\gamma \tilde{\mu} g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{i2}\right] -$$

$$\frac{1}{2} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{i2}\right] -$$

$$\frac{1}{2} m_1 s_\gamma \tilde{\mu} g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{i2}\right] -$$

$$\frac{1}{4} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{p}\right] +$$

$$\frac{1}{4} s_\gamma \tilde{\mu} c_\gamma g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \left(m_1 c_\gamma + s_\gamma \tilde{\mu}\right) \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{p}\right] -$$

$$\frac{1}{4} s_\gamma \tilde{\mu} c_\gamma g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \left(m_1 c_\gamma + s_\gamma \tilde{\mu}\right) \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{p}\right] -$$

$$\frac{3}{8} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i1}\right] -$$

$$\frac{1}{4} m_1 s_\gamma \tilde{\mu} g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i1}\right] +$$

$$\frac{1}{4} g_1^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i1}\right] +$$

$$\frac{1}{4} m_1 s_\gamma \tilde{\mu} g_1^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i1}\right] -$$

$$c_\gamma g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{3,2,1,-3}\left[\tilde{\mu}, m_2, m_1\right] -$$

$$c_\gamma g_1^2 g_2^2 y_e^{i1i2} \left(m_1 m_2 + 4 s_\gamma \tilde{\mu} c_\gamma \left(m_1 + m_2\right) + 4 s_\gamma^2 \tilde{\mu}^2 c_\gamma^2\right) \mathrm{LF}_{3,2,1,-2}\left[\tilde{\mu}, m_2, m_1\right] -$$

$$4 m_1 m_2 g_1^2 g_2^2 s_\gamma^2 \tilde{\mu}^2 c_\gamma^3 y_e^{i1i2} \mathrm{LF}_{3,2,1,-1}\left[\tilde{\mu}, m_2, m_1\right] +$$

$$\frac{3}{8} g_2^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{e}}^{r}\right] +$$

$$\frac{1}{4} s_\gamma \tilde{\mu} c_\gamma g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \left(2 m_2 c_\gamma + s_\gamma \tilde{\mu}\right) \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{e}}^{r}\right] -$$

$$\frac{1}{4} g_2^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}\left[\tilde{\mu}, m_2, m_{\tilde{e}}^{r}\right] -$$

$$\frac{1}{4} s_\gamma \tilde{\mu} c_\gamma g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \left(2 m_2 c_\gamma + s_\gamma \tilde{\mu}\right) \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{e}}^{r}\right] +$$

$$3 g_2^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{p}\right] +$$

$$\frac{3}{4} s_\gamma \tilde{\mu} c_\gamma g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \left(3 m_2 c_\gamma + s_\gamma \tilde{\mu}\right) \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{p}\right] -$$

$$\frac{3}{2} g_2^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{p}\right] -$$

$$\frac{3}{4} s_\gamma \tilde{\mu} c_\gamma g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \left(3 m_2 c_\gamma + s_\gamma \tilde{\mu}\right) \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{p}\right] +$$

$$\frac{9}{8} g_2^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i1}\right] +$$

$$\frac{3}{4} m_2 s_\gamma \tilde{\mu} g_2^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i1}\right] -$$

$$\frac{3}{4} g_2^2 c_\gamma^3 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i1}\right] -$$

$$\frac{3}{4} m_2 s_\gamma \tilde{\mu} g_2^2 c_\gamma^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i1}\right] -$$

$$\frac{1}{4} c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}\right] -$$

$$\frac{1}{4} c_\gamma^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{si2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}\right] -$$

$$\frac{3}{8} c_\gamma g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{1,1,1,1,-1}\left[m_1, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] -$$

$$\frac{1}{4} c_\gamma g_1^2 g_2^2 y_e^{i1i2} \left(m_1 m_2 + 2 s_\gamma \tilde{\mu} c_\gamma \left(m_1 + m_2\right)\right) \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] +$$

$$\frac{1}{8} c_\gamma g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{2,1,1,1,-2}\left[m_1, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] +$$

$$\frac{1}{8} c_\gamma g_1^2 g_2^2 y_e^{i1i2} \left(m_1 m_2 + 2 s_\gamma \tilde{\mu} c_\gamma \left(m_1 + m_2\right)\right) \mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] -$$

$$\frac{1}{8} c_\gamma g_1^2 y_e^{pi2} y_e^{i1r} \left(m_1 c_\gamma + 3 s_\gamma \tilde{\mu}\right) \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}\right) \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}\right] +$$

$$\frac{3}{16} c_\gamma g_1^2 y_e^{pi2} y_e^{i1r} \left(m_1 c_\gamma + s_\gamma \tilde{\mu}\right) \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}\right) \mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}\right] +$$

$$\frac{1}{4} c_\gamma g_1^2 y_e^{pi2} \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}\right) \left(c_\gamma a_e^{i1r} - s_\gamma \tilde{\mu} y_e^{i1r}\right) \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}\right] -$$

$$\frac{1}{8} c_\gamma g_1^2 y_e^{pi2} \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}\right) \left(c_\gamma a_e^{i1r} - s_\gamma \tilde{\mu} y_e^{i1r}\right) \mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}\right] +$$

$$\frac{1}{8} c_\gamma g_1^2 \overline{y_e}^{pr} y_e^{pi2} \left(3 m_1 c_\gamma + s_\gamma \tilde{\mu}\right) \left(c_\gamma a_e^{i1r} - s_\gamma \tilde{\mu} y_e^{i1r}\right) \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] -$$

$$\frac{1}{16} c_\gamma g_1^2 \overline{y_e}^{pr} y_e^{pi2} \left(m_1 c_\gamma + s_\gamma \tilde{\mu}\right) \left(c_\gamma a_e^{i1r} - s_\gamma \tilde{\mu} y_e^{i1r}\right) \mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] +$$

$$c_\gamma g_1^2 y_e^{i1r} \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}\right) \left(c_\gamma a_e^{pi2} - s_\gamma \tilde{\mu} y_e^{pi2}\right) \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right] -$$

$$\frac{1}{2} c_\gamma g_1{}^2 y_e{}^{i1r} (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) (c_\gamma a_e{}^{pi2} - s_\gamma \tilde{\mu} y_e{}^{pi2}) \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{e}}{}^{i2}, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{p}] +$$

$$m_1 g_1{}^2 c_\gamma{}^2 \overline{y_e}^{pr} y_e{}^{pi2} (-c_\gamma a_e{}^{i1r} + s_\gamma \tilde{\mu} y_e{}^{i1r}) \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{e}}{}^{i2}, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{i1}] -$$

$$g_1{}^2 c_\gamma{}^3 \overline{y_e}^{pr} y_e{}^{pi2} y_e{}^{i1r} \mathrm{LF}_{1,1,1,1,-1}[m_1, m_{\tilde{e}}{}^{i2}, m_{\tilde{e}}{}^{r}, \tilde{\mu}] -$$

$$m_1 s_\gamma \tilde{\mu} g_1{}^2 c_\gamma{}^2 \overline{y_e}^{pr} y_e{}^{pi2} y_e{}^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{e}}{}^{i2}, m_{\tilde{e}}{}^{r}, \tilde{\mu}] -$$

$$\frac{1}{4} c_\gamma g_1{}^2 \overline{y_e}^{pr} y_e{}^{i1r} (3 m_1 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma a_e{}^{pi2} - s_\gamma \tilde{\mu} y_e{}^{pi2}) \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{e}}{}^{i2}, m_{\tilde{l}}{}^{p}, \tilde{\mu}] +$$

$$\frac{1}{8} c_\gamma g_1{}^2 \overline{y_e}^{pr} y_e{}^{i1r} (m_1 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma a_e{}^{pi2} - s_\gamma \tilde{\mu} y_e{}^{pi2}) \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{e}}{}^{i2}, m_{\tilde{l}}{}^{p}, \tilde{\mu}] +$$

$$m_1 g_1{}^2 c_\gamma{}^2 \overline{y_e}^{pr} y_e{}^{i1r} (-c_\gamma a_e{}^{pi2} + s_\gamma \tilde{\mu} y_e{}^{pi2}) \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{e}}{}^{i2}, m_{\tilde{l}}{}^{i1}, m_{\tilde{l}}{}^{p}] -$$

$$g_1{}^4 (m_1 + s_\gamma \tilde{\mu} c_\gamma) (c_\gamma a_e{}^{i1i2} - s_\gamma \tilde{\mu} y_e{}^{i1i2}) \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{e}}{}^{i2}, m_{\tilde{l}}{}^{i1}, \tilde{\mu}] +$$

$$s_\gamma \tilde{\mu} c_\gamma g_1{}^4 m_1{}^2 (-c_\gamma a_e{}^{i1i2} + s_\gamma \tilde{\mu} y_e{}^{i1i2}) \mathrm{LF}_{2,1,1,1,0}[m_1, m_{\tilde{e}}{}^{i2}, m_{\tilde{l}}{}^{i1}, \tilde{\mu}] +$$

$$\frac{1}{2} g_1{}^2 c_\gamma{}^3 \overline{y_e}^{pr} y_e{}^{pi2} y_e{}^{i1r} \mathrm{LF}_{1,1,1,1,-1}[m_1, m_{\tilde{l}}{}^{i1}, m_{\tilde{l}}{}^{p}, \tilde{\mu}] +$$

$$\frac{1}{2} m_1 s_\gamma \tilde{\mu} g_1{}^2 c_\gamma{}^2 \overline{y_e}^{pr} y_e{}^{pi2} y_e{}^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{l}}{}^{i1}, m_{\tilde{l}}{}^{p}, \tilde{\mu}] +$$

$$\frac{1}{8} c_\gamma g_1{}^2 g_2{}^2 y_e{}^{i1i2} \mathrm{LF}_{2,1,1,1,-2}[m_2, m_1, m_{\tilde{l}}{}^{i1}, \tilde{\mu}] +$$

$$\frac{1}{8} c_\gamma g_1{}^2 g_2{}^2 y_e{}^{i1i2} (m_1 m_2 + 2 s_\gamma \tilde{\mu} c_\gamma (m_1 + m_2)) \mathrm{LF}_{2,1,1,1,-1}[m_2, m_1, m_{\tilde{l}}{}^{i1}, \tilde{\mu}] -$$

$$\frac{1}{8} c_\gamma g_2{}^2 y_e{}^{pi2} y_e{}^{i1r} (m_2 c_\gamma + 3 s_\gamma \tilde{\mu}) (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{p}, \tilde{\mu}] +$$

$$\frac{3}{16} c_\gamma g_2{}^2 y_e{}^{pi2} y_e{}^{i1r} (m_2 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{p}, \tilde{\mu}] +$$

$$\frac{1}{4} c_\gamma g_2{}^2 y_e{}^{pi2} (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) (c_\gamma a_e{}^{i1r} - s_\gamma \tilde{\mu} y_e{}^{i1r}) \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{i1}, m_{\tilde{l}}{}^{p}] -$$

$$\frac{1}{8} c_\gamma g_2{}^2 y_e{}^{pi2} (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) (c_\gamma a_e{}^{i1r} - s_\gamma \tilde{\mu} y_e{}^{i1r}) \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{i1}, m_{\tilde{l}}{}^{p}] -$$

$$\frac{1}{8} c_\gamma g_2{}^2 \overline{y_e}^{pr} y_e{}^{pi2} (3 m_2 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma a_e{}^{i1r} - s_\gamma \tilde{\mu} y_e{}^{i1r}) \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{i1}, \tilde{\mu}] +$$

$$\frac{1}{16} c_\gamma g_2{}^2 \overline{y_e}^{pr} y_e{}^{pi2} (m_2 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma a_e{}^{i1r} - s_\gamma \tilde{\mu} y_e{}^{i1r}) \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{i1}, \tilde{\mu}] -$$

$$\frac{1}{2} g_2{}^2 c_\gamma{}^3 \overline{y_e}^{pr} y_e{}^{pi2} y_e{}^{i1r} \mathrm{LF}_{1,1,1,1,-1}[m_2, m_{\tilde{l}}{}^{i1}, m_{\tilde{l}}{}^{p}, \tilde{\mu}] -$$

$$\frac{1}{2} m_2 s_\gamma \tilde{\mu} g_2{}^2 c_\gamma{}^2 \overline{y_e}^{pr} y_e{}^{pi2} y_e{}^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{l}}{}^{i1}, m_{\tilde{l}}{}^{p}, \tilde{\mu}] -$$

$$3 s_\gamma \frac{1}{m_\Phi{}^2} y_e{}^{i1i2} (-s_\gamma c_\gamma \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} (s_\gamma{}^2 \tilde{\mu}^2 y_d{}^{pt} y_d{}^{sr} + c_\gamma a_d{}^{pt} (c_\gamma a_d{}^{sr} - 2 s_\gamma \tilde{\mu} y_d{}^{sr})) +$$

$$\overline{a_d}^{pr} (\tilde{\mu} \overline{y_d}^{st} (c_\gamma{}^2 - s_\gamma{}^2) (c_\gamma a_d{}^{pt} - s_\gamma \tilde{\mu} y_d{}^{pt}) (c_\gamma a_d{}^{sr} - s_\gamma \tilde{\mu} y_d{}^{sr}) + s_\gamma c_\gamma \overline{a_d}^{st}$$

$$(s_\gamma{}^2 \tilde{\mu}^2 y_d{}^{pt} y_d{}^{sr} + c_\gamma a_d{}^{pt} (c_\gamma a_d{}^{sr} - 2 s_\gamma \tilde{\mu} y_d{}^{sr})))) \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}{}^{r}, m_{\tilde{d}}{}^{t}, m_{\tilde{q}}{}^{p}, m_{\tilde{q}}{}^{s}] -$$

$$\frac{3}{4} c_\gamma y_e{}^{i1i2} (s_\gamma{}^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} (s_\gamma{}^2 \tilde{\mu}^2 y_d{}^{pt} y_d{}^{sr} + c_\gamma a_d{}^{pt} (c_\gamma a_d{}^{sr} - s_\gamma \tilde{\mu} y_d{}^{sr})) +$$

$$c_\gamma \overline{a_d}^{pr} (-s_\gamma \tilde{\mu} \overline{y_d}^{st} (-c_\gamma a_d{}^{pt} + s_\gamma \tilde{\mu} y_d{}^{pt}) (-c_\gamma a_d{}^{sr} + s_\gamma \tilde{\mu} y_d{}^{sr}) + c_\gamma \overline{a_d}^{st}$$

$$(s_\gamma{}^2 \tilde{\mu}^2 y_d{}^{pt} y_d{}^{sr} + c_\gamma a_d{}^{pt} (c_\gamma a_d{}^{sr} - s_\gamma \tilde{\mu} y_d{}^{sr})))) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}{}^{r}, m_{\tilde{q}}{}^{p}, m_{\tilde{d}}{}^{t}, m_{\tilde{q}}{}^{s}] +$$

$$\frac{3}{4} c_\gamma y_e{}^{i1i2} (s_\gamma{}^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} (s_\gamma{}^2 \tilde{\mu}^2 y_d{}^{pt} y_d{}^{sr} + c_\gamma a_d{}^{pt} (c_\gamma a_d{}^{sr} - s_\gamma \tilde{\mu} y_d{}^{sr})) +$$

$$c_\gamma \overline{a_d}^{pr} (-s_\gamma \tilde{\mu} \overline{y_d}^{st} (-c_\gamma a_d{}^{pt} + s_\gamma \tilde{\mu} y_d{}^{pt}) (-c_\gamma a_d{}^{sr} + s_\gamma \tilde{\mu} y_d{}^{sr}) +$$

$$c_\gamma \overline{a_d}^{st} (s_\gamma{}^2 \tilde{\mu}^2 y_d{}^{pt} y_d{}^{sr} + c_\gamma a_d{}^{pt} (c_\gamma a_d{}^{sr} - s_\gamma \tilde{\mu} y_d{}^{sr}))))$$

$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}{}^{r}, m_{\tilde{q}}{}^{s}, m_{\tilde{d}}{}^{t}, m_{\tilde{q}}{}^{p}] - \frac{3}{4} s_\gamma \tilde{\mu} c_\gamma{}^2 y_e{}^{i1i2} (s_\gamma{}^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d{}^{sr} a_d{}^{pt} +$$

$$\overline{a_d}^{pr} (c_\gamma{}^2 y_d{}^{sr} \overline{a_d}^{st} a_d{}^{pt} + \overline{y_d}^{st} (c_\gamma a_d{}^{pt} - s_\gamma \tilde{\mu} y_d{}^{pt}) (c_\gamma a_d{}^{sr} - s_\gamma \tilde{\mu} y_d{}^{sr})))$$

$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}{}^{t}, m_{\tilde{q}}{}^{p}, m_{\tilde{d}}{}^{r}, m_{\tilde{q}}{}^{s}] + \frac{3}{4} s_\gamma \tilde{\mu} c_\gamma{}^2 y_e{}^{i1i2} (s_\gamma{}^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d{}^{sr} a_d{}^{pt} +$$

$$\overline{a_d}^{pr} (c_\gamma{}^2 y_d{}^{sr} \overline{a_d}^{st} a_d{}^{pt} + \overline{y_d}^{st} (c_\gamma a_d{}^{pt} - s_\gamma \tilde{\mu} y_d{}^{pt}) (c_\gamma a_d{}^{sr} - s_\gamma \tilde{\mu} y_d{}^{sr})))$$

$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}{}^{t}, m_{\tilde{q}}{}^{s}, m_{\tilde{d}}{}^{r}, m_{\tilde{q}}{}^{p}] + \frac{1}{2} c_\gamma g_1{}^2 y_e{}^{i1r} (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr})$$

$$(c_\gamma a_e{}^{pi2} - s_\gamma \tilde{\mu} y_e{}^{pi2}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}{}^{r}, m_1, m_{\tilde{e}}{}^{i2}, m_{\tilde{l}}{}^{p}] -$$

$$\frac{3}{16} c_\gamma g_1{}^2 y_e{}^{pi2} y_e{}^{i1r} (m_1 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}{}^{r}, m_1, m_{\tilde{l}}{}^{p}, \tilde{\mu}] +$$

$$\frac{1}{16} c_\gamma g_1{}^2 \overline{y_e}^{pr} y_e{}^{pi2} (m_1 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma a_e{}^{i1r} - s_\gamma \tilde{\mu} y_e{}^{i1r}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}{}^{r}, m_1, m_{\tilde{l}}{}^{i1}, \tilde{\mu}] +$$

$$\frac{1}{16} c_\gamma g_2{}^2 y_e{}^{pi2} y_e{}^{i1r} (m_2 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}{}^{r}, m_2, m_{\tilde{l}}{}^{p}, \tilde{\mu}] -$$

$$\frac{1}{16} c_\gamma g_2{}^2 \overline{y_e}^{pr} y_e{}^{pi2} (m_2 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma a_e{}^{i1r} - s_\gamma \tilde{\mu} y_e{}^{i1r}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}{}^{r}, m_2, m_{\tilde{l}}{}^{i1}, \tilde{\mu}] +$$

$$s_\gamma \frac{1}{m_\Phi{}^2} y_e{}^{i1i2} (s_\gamma c_\gamma \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} (s_\gamma{}^2 \tilde{\mu}^2 y_e{}^{pt} y_e{}^{sr} + c_\gamma a_e{}^{pt} (c_\gamma a_e{}^{sr} - 2 s_\gamma \tilde{\mu} y_e{}^{sr})) -$$

$$\overline{a_e}^{pr} (\tilde{\mu} \overline{y_e}^{st} (c_\gamma{}^2 - s_\gamma{}^2) (c_\gamma a_e{}^{pt} - s_\gamma \tilde{\mu} y_e{}^{pt}) (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr}) + s_\gamma c_\gamma \overline{a_e}^{st}$$

$$(s_\gamma{}^2 \tilde{\mu}^2 y_e{}^{pt} y_e{}^{sr} + c_\gamma a_e{}^{pt} (c_\gamma a_e{}^{sr} - 2 s_\gamma \tilde{\mu} y_e{}^{sr})))) \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{e}}{}^{r}, m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{p}, m_{\tilde{l}}{}^{s}] +$$

$$\frac{1}{2} c_\gamma{}^2 \overline{y_e}^{st} y_e{}^{si2} y_e{}^{i1r} \overline{a_e}^{pr} (c_\gamma a_e{}^{pt} - s_\gamma \tilde{\mu} y_e{}^{pt}) \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{e}}{}^{r}, m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{p}, \tilde{\mu}] +$$

$$\frac{1}{4} c_\gamma{}^2 \overline{y_e}^{st} y_e{}^{si2} y_e{}^{i1r} \overline{a_e}^{pr} (-c_\gamma a_e{}^{pt} + s_\gamma \tilde{\mu} y_e{}^{pt}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}{}^{r}, m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{p}, \tilde{\mu}] +$$

$$\frac{1}{2} s_\gamma \tilde{\mu} c_\gamma \overline{y_e}^{pr} \overline{y_e}^{st} y_e{}^{pi2} y_e{}^{i1t} (-c_\gamma a_e{}^{sr} + s_\gamma \tilde{\mu} y_e{}^{sr}) \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{e}}{}^{r}, m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{s}, \tilde{\mu}] +$$

$$\frac{1}{4} s_\gamma \tilde{\mu} c_\gamma \overline{y_e}^{pr} \overline{y_e}^{st} y_e{}^{pi2} y_e{}^{i1t} (-c_\gamma a_e{}^{sr} + s_\gamma \tilde{\mu} y_e{}^{sr}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}{}^{r}, m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{s}, \tilde{\mu}] -$$

$$\frac{1}{4} c_\gamma y_e{}^{i1i2} (s_\gamma{}^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} (s_\gamma{}^2 \tilde{\mu}^2 y_e{}^{pt} y_e{}^{sr} + c_\gamma a_e{}^{pt} (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr})) +$$

$$c_\gamma \overline{a_e}^{pr} (-s_\gamma \tilde{\mu} \overline{y_e}^{st} (-c_\gamma a_e{}^{pt} + s_\gamma \tilde{\mu} y_e{}^{pt}) (-c_\gamma a_e{}^{sr} + s_\gamma \tilde{\mu} y_e{}^{sr}) + c_\gamma \overline{a_e}^{st}$$

$$(s_\gamma{}^2 \tilde{\mu}^2 y_e{}^{pt} y_e{}^{sr} + c_\gamma a_e{}^{pt} (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr})))) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{p}, m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{s}] +$$

$$\frac{1}{2} c_\gamma{}^2 \overline{y_e}^{st} y_e{}^{pi2} y_e{}^{i1t} \overline{a_e}^{pr} (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr}) \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{p}, m_{\tilde{l}}{}^{s}, \tilde{\mu}] +$$

$$\frac{1}{4} c_\gamma y_e{}^{i1i2} (s_\gamma{}^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} (s_\gamma{}^2 \tilde{\mu}^2 y_e{}^{pt} y_e{}^{sr} + c_\gamma a_e{}^{pt} (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr})) +$$

$$c_\gamma \overline{a_e}^{pr} (-s_\gamma \tilde{\mu} \overline{y_e}^{st} (-c_\gamma a_e{}^{pt} + s_\gamma \tilde{\mu} y_e{}^{pt}) (-c_\gamma a_e{}^{sr} + s_\gamma \tilde{\mu} y_e{}^{sr}) + c_\gamma \overline{a_e}^{st}$$

$$(s_\gamma{}^2 \tilde{\mu}^2 y_e{}^{pt} y_e{}^{sr} + c_\gamma a_e{}^{pt} (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr})))) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{s}, m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{p}] +$$

$$\frac{1}{4} c_\gamma{}^2 \overline{y_e}^{st} y_e{}^{si2} y_e{}^{i1r} \overline{a_e}^{pr} (c_\gamma a_e{}^{pt} - s_\gamma \tilde{\mu} y_e{}^{pt}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}{}^{t}, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{p}, \tilde{\mu}] +$$

$$\frac{1}{4} s_\gamma \tilde{\mu} c_\gamma \overline{y_e}^{pr} \overline{y_e}^{st} y_e{}^{pi2} y_e{}^{i1t} (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}{}^{t}, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{s}, \tilde{\mu}] -$$

$$\frac{1}{4} s_\gamma \tilde{\mu} c_\gamma{}^2 y_e{}^{i1i2} (s_\gamma{}^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e{}^{sr} a_e{}^{pt} +$$

$$\overline{a_e}^{pr} (c_\gamma{}^2 y_e{}^{sr} \overline{a_e}^{st} a_e{}^{pt} + \overline{y_e}^{st} (c_\gamma a_e{}^{pt} - s_\gamma \tilde{\mu} y_e{}^{pt}) (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr})))$$

$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{p}, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{s}] + \frac{1}{2} s_\gamma \tilde{\mu} c_\gamma \overline{y_e}^{pr} \overline{y_e}^{st} y_e{}^{si2} y_e{}^{i1r} (-c_\gamma a_e{}^{pt} + s_\gamma \tilde{\mu} y_e{}^{pt})$$

$$\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{p}, m_{\tilde{l}}{}^{s}, \tilde{\mu}] + \frac{1}{4} s_\gamma \tilde{\mu} c_\gamma{}^2 y_e{}^{i1i2} (s_\gamma{}^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e{}^{sr} a_e{}^{pt} +$$

$$\overline{a_e}^{pr} (c_\gamma{}^2 y_e{}^{sr} \overline{a_e}^{st} a_e{}^{pt} + \overline{y_e}^{st} (c_\gamma a_e{}^{pt} - s_\gamma \tilde{\mu} y_e{}^{pt}) (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr})))$$

$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{s}, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{p}] - \frac{1}{2} c_\gamma g_1{}^2 y_e{}^{i1r} (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr})$$

$$(c_\gamma a_e{}^{pi2} - s_\gamma \tilde{\mu} y_e{}^{pi2}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}{}^{i2}, m_1, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{p}] +$$

$$\frac{1}{8} c_\gamma g_1{}^2 \overline{y_e}^{pr} y_e{}^{i1r} (m_1 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma a_e{}^{pi2} - s_\gamma \tilde{\mu} y_e{}^{pi2}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}{}^{i2}, m_1, m_{\tilde{l}}{}^{p}, \tilde{\mu}] +$$

$$\frac{1}{8} c_\gamma g_1{}^2 y_e{}^{pi2} (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) (c_\gamma a_e{}^{i1r} - s_\gamma \tilde{\mu} y_e{}^{i1r}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}{}^{p}, m_1, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{i1}] +$$

$$\frac{3}{16} c_\gamma g_1{}^2 y_e{}^{pi2} y_e{}^{i1r} (m_1 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}{}^{p}, m_1, m_{\tilde{e}}{}^{r}, \tilde{\mu}] -$$

$$\frac{1}{8} c_\gamma g_1{}^2 \overline{y_e}^{pr} y_e{}^{i1r} (m_1 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma a_e{}^{pi2} - s_\gamma \tilde{\mu} y_e{}^{pi2}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}{}^{p}, m_1, m_{\tilde{e}}{}^{i2}, \tilde{\mu}] +$$

$$\frac{1}{8} c_\gamma g_2{}^2 y_e{}^{pi2} (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) (c_\gamma a_e{}^{i1r} - s_\gamma \tilde{\mu} y_e{}^{i1r}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}{}^{p}, m_2, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{i1}] -$$

$$\frac{1}{16} c_\gamma g_2{}^2 y_e{}^{pi2} y_e{}^{i1r} (m_2 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}{}^{p}, m_2, m_{\tilde{e}}{}^{r}, \tilde{\mu}] -$$

$$\frac{1}{2} c_\gamma y_e{}^{i1i2} (s_\gamma{}^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} (s_\gamma{}^2 \tilde{\mu}^2 y_e{}^{pt} y_e{}^{sr} + c_\gamma a_e{}^{pt} (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr})) +$$

$$c_\gamma \overline{a_e}^{pr} (-s_\gamma \tilde{\mu} \overline{y_e}^{st} (-c_\gamma a_e{}^{pt} + s_\gamma \tilde{\mu} y_e{}^{pt}) (-c_\gamma a_e{}^{sr} + s_\gamma \tilde{\mu} y_e{}^{sr}) + c_\gamma \overline{a_e}^{st}$$

$$(s_\gamma{}^2 \tilde{\mu}^2 y_e{}^{pt} y_e{}^{sr} + c_\gamma a_e{}^{pt} (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr})))) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{l}}{}^{p}, m_{\tilde{e}}{}^{r}, m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{s}] +$$

$$\frac{1}{2} c_\gamma y_e{}^{i1i2} (s_\gamma{}^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} (s_\gamma{}^2 \tilde{\mu}^2 y_e{}^{pt} y_e{}^{sr} + c_\gamma a_e{}^{pt} (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr})) +$$

$$c_\gamma \overline{a_e}^{pr} (-s_\gamma \tilde{\mu} \overline{y_e}^{st} (-c_\gamma a_e{}^{pt} + s_\gamma \tilde{\mu} y_e{}^{pt}) (-c_\gamma a_e{}^{sr} + s_\gamma \tilde{\mu} y_e{}^{sr}) + c_\gamma \overline{a_e}^{st}$$

$$(s_\gamma{}^2 \tilde{\mu}^2 y_e{}^{pt} y_e{}^{sr} + c_\gamma a_e{}^{pt} (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr})))) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{l}}{}^{p}, m_{\tilde{e}}{}^{r}, m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{s}] -$$

$$\frac{1}{4} c_\gamma \overline{y_e}^{st} y_e{}^{pi2} y_e{}^{i1t} (s_\gamma{}^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e{}^{sr} + c_\gamma \overline{a_e}^{pr} (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr}))$$

$$\mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}{}^{p}, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{s}, \tilde{\mu}] +$$

$$\frac{1}{4} s_\gamma \tilde{\mu} c_\gamma \overline{y_e}^{pr} \overline{y_e}^{st} y_e{}^{si2} y_e{}^{i1r} (-c_\gamma a_e{}^{pt} + s_\gamma \tilde{\mu} y_e{}^{pt}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}{}^{p}, m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{s}, \tilde{\mu}] +$$

$$\frac{1}{2} s_\gamma \tilde{\mu} c_\gamma{}^2 y_e{}^{i1i2} (s_\gamma{}^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e{}^{sr} a_e{}^{pt} +$$

$$\overline{a_e}^{pr} (c_\gamma{}^2 y_e{}^{sr} \overline{a_e}^{st} a_e{}^{pt} + \overline{y_e}^{st} (c_\gamma a_e{}^{pt} - s_\gamma \tilde{\mu} y_e{}^{pt}) (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr})))$$

$$\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{l}}{}^{s}, m_{\tilde{e}}{}^{r}, m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{p}] - \frac{1}{2} s_\gamma \tilde{\mu} c_\gamma{}^2 y_e{}^{i1i2} (s_\gamma{}^2 \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e{}^{sr} a_e{}^{pt} +$$

$$\overline{a_e}^{pr} (c_\gamma{}^2 y_e{}^{sr} \overline{a_e}^{st} a_e{}^{pt} + \overline{y_e}^{st} (c_\gamma a_e{}^{pt} - s_\gamma \tilde{\mu} y_e{}^{pt}) (c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr})))$$

$$\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{l}}{}^{s}, m_{\tilde{e}}{}^{r}, m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{p}] + \frac{1}{4} c_\gamma{}^2 \overline{y_e}^{st} y_e{}^{pi2} y_e{}^{i1t} \overline{a_e}^{pr}$$

$$(c_\gamma a_e{}^{sr} - s_\gamma \tilde{\mu} y_e{}^{sr}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}{}^{s}, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{p}, \tilde{\mu}] +$$

$$\frac{1}{4} s_\gamma \tilde{\mu} c_\gamma{}^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e{}^{si2} y_e{}^{i1r} a_e{}^{pt} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}{}^{s}, m_{\tilde{e}}{}^{t}, m_{\tilde{l}}{}^{p}, \tilde{\mu}] -$$

$$\frac{1}{8} c_\gamma g_1{}^2 y_e{}^{pi2} (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) (c_\gamma a_e{}^{i1r} - s_\gamma \tilde{\mu} y_e{}^{i1r}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}{}^{i1}, m_1, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{p}] -$$

$$\frac{1}{16} c_\gamma g_1{}^2 \overline{y_e}^{pr} y_e{}^{pi2} (m_1 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma a_e{}^{i1r} - s_\gamma \tilde{\mu} y_e{}^{i1r}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}{}^{i1}, m_1, m_{\tilde{e}}{}^{r}, \tilde{\mu}] -$$

$$\frac{1}{8} c_\gamma g_2{}^2 y_e{}^{pi2} (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) (c_\gamma a_e{}^{i1r} - s_\gamma \tilde{\mu} y_e{}^{i1r}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}{}^{i1}, m_2, m_{\tilde{e}}{}^{r}, m_{\tilde{l}}{}^{p}] +$$

$$\frac{1}{16} c_\gamma g_2{}^2 \overline{y_e}^{pr} y_e{}^{pi2} (m_2 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma a_e{}^{i1r} - s_\gamma \tilde{\mu} y_e{}^{i1r}) \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}{}^{i1}, m_2, m_{\tilde{e}}{}^{r}, \tilde{\mu}] -$$

$$\frac{3}{2} c_\gamma y_e{}^{i1i2} (s_\gamma{}^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} (s_\gamma{}^2 \tilde{\mu}^2 y_d{}^{pt} y_d{}^{sr} + c_\gamma a_d{}^{pt} (c_\gamma a_d{}^{sr} - s_\gamma \tilde{\mu} y_d{}^{sr})) +$$

$$c_\gamma \overline{a_d}^{pr} (-s_\gamma \tilde{\mu} \overline{y_d}^{st} (-c_\gamma a_d{}^{pt} + s_\gamma \tilde{\mu} y_d{}^{pt}) (-c_\gamma a_d{}^{sr} + s_\gamma \tilde{\mu} y_d{}^{sr}) + c_\gamma \overline{a_d}^{st}$$

$$(s_\gamma{}^2 \tilde{\mu}^2 y_d{}^{pt} y_d{}^{sr} + c_\gamma a_d{}^{pt} (c_\gamma a_d{}^{sr} - s_\gamma \tilde{\mu} y_d{}^{sr})))) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}{}^{p}, m_{\tilde{d}}{}^{r}, m_{\tilde{d}}{}^{t}, m_{\tilde{q}}{}^{s}] +$$

$$\frac{3}{2} c_\gamma y_e{}^{i1i2} (s_\gamma{}^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} (s_\gamma{}^2 \tilde{\mu}^2 y_d{}^{pt} y_d{}^{sr} + c_\gamma a_d{}^{pt} (c_\gamma a_d{}^{sr} - s_\gamma \tilde{\mu} y_d{}^{sr})) +$$

$$c_\gamma \overline{a_d}^{pr} (-s_\gamma \tilde{\mu} \overline{y_d}^{st} (-c_\gamma a_d{}^{pt} + s_\gamma \tilde{\mu} y_d{}^{pt}) (-c_\gamma a_d{}^{sr} + s_\gamma \tilde{\mu} y_d{}^{sr}) +$$

$$c_\gamma \overline{a_d}^{st} (s_\gamma{}^2 \tilde{\mu}^2 y_d{}^{pt} y_d{}^{sr} + c_\gamma a_d{}^{pt} (c_\gamma a_d{}^{sr} - s_\gamma \tilde{\mu} y_d{}^{sr}))))$$

$$\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}{}^{p}, m_{\tilde{d}}{}^{r}, m_{\tilde{d}}{}^{t}, m_{\tilde{q}}{}^{s}] + \frac{3}{2} c_\gamma{}^2 y_e{}^{i1i2} (c_\gamma a_d{}^{sr} - s_\gamma \tilde{\mu} y_d{}^{sr}) (s_\gamma a_u{}^{pt} - \tilde{\mu} c_\gamma y_u{}^{pt})$$

$$(s_\gamma \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_u}^{st} + \overline{a_d}^{pr} (s_\gamma \overline{a_u}^{st} - \tilde{\mu} c_\gamma \overline{y_u}^{st})) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}{}^{p}, m_{\tilde{q}}{}^{s}, m_{\tilde{d}}{}^{r}, m_{\tilde{u}}{}^{t}] +$$

$$\frac{3}{2} \tilde{\mu} c_\gamma s_\gamma{}^2 \overline{y_d}^{st} y_e{}^{i1i2} \overline{a_u}^{pr} (c_\gamma a_d{}^{pt} - s_\gamma \tilde{\mu} y_d{}^{pt}) (-s_\gamma a_u{}^{sr} + \tilde{\mu} c_\gamma y_u{}^{sr})$$

$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}{}^{p}, m_{\tilde{q}}{}^{s}, m_{\tilde{d}}{}^{t}, m_{\tilde{u}}{}^{r}] + 3 s_\gamma \frac{1}{m_\Phi{}^2} y_e{}^{i1i2} (\tilde{\mu}^2 c_\gamma{}^2 \overline{y_u}^{pr} \overline{y_u}^{st}$$

$$(-s_\gamma c_\gamma \tilde{\mu}^2 y_u{}^{pt} y_u{}^{sr} + a_u{}^{pt} (s_\gamma c_\gamma a_u{}^{sr} + \tilde{\mu} y_u{}^{sr} (-c_\gamma{}^2 + s_\gamma{}^2))) + s_\gamma \overline{a_u}^{pr} (\tilde{\mu} c_\gamma \overline{y_u}^{st}$$

$$(\tilde{\mu} y_u{}^{pt} (a_u{}^{sr} (c_\gamma{}^2 - s_\gamma{}^2) + 2 s_\gamma \tilde{\mu} c_\gamma y_u{}^{sr}) + a_u{}^{pt} (-2 s_\gamma c_\gamma a_u{}^{sr} + \tilde{\mu} y_u{}^{sr} (c_\gamma{}^2 - s_\gamma{}^2))) +$$

$$s_\gamma \overline{a_u}^{st} (-s_\gamma c_\gamma \tilde{\mu}^2 y_u{}^{pt} y_u{}^{sr} + a_u{}^{pt} (s_\gamma c_\gamma a_u{}^{sr} + \tilde{\mu} y_u{}^{sr} (-c_\gamma{}^2 + s_\gamma{}^2))))) \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{q}}{}^{p},$$

$$m_{\tilde{q}}{}^{s}, m_{\tilde{u}}{}^{r}, m_{\tilde{u}}{}^{t}] - \frac{3}{2} c_\gamma y_e{}^{i1i2} (\tilde{\mu}^2 c_\gamma{}^2 \overline{y_u}^{pr} \overline{y_u}^{st} (\tilde{\mu}^2 c_\gamma{}^2 y_u{}^{pt} y_u{}^{sr} + s_\gamma a_u{}^{pt} (s_\gamma a_u{}^{sr} - \tilde{\mu} c_\gamma y_u{}^{sr})) +$$

$$s_\gamma \overline{a_u}^{pr} (-\tilde{\mu} c_\gamma \overline{y_u}^{st} (-s_\gamma a_u{}^{pt} + \tilde{\mu} c_\gamma y_u{}^{pt}) (-s_\gamma a_u{}^{sr} + \tilde{\mu} c_\gamma y_u{}^{sr}) + s_\gamma \overline{a_u}^{st}$$

$$(\tilde{\mu}^2 c_\gamma{}^2 y_u{}^{pt} y_u{}^{sr} + s_\gamma a_u{}^{pt} (s_\gamma a_u{}^{sr} - \tilde{\mu} c_\gamma y_u{}^{sr})))) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}{}^{p}, m_{\tilde{q}}{}^{s}, m_{\tilde{u}}{}^{r}, m_{\tilde{u}}{}^{t}] +$$

$$\frac{3}{2} c_\gamma y_e{}^{i1i2} (\tilde{\mu}^2 c_\gamma{}^2 \overline{y_u}^{pr} \overline{y_u}^{st} (\tilde{\mu}^2 c_\gamma{}^2 y_u{}^{pt} y_u{}^{sr} + s_\gamma a_u{}^{pt} (s_\gamma a_u{}^{sr} - \tilde{\mu} c_\gamma y_u{}^{sr})) +$$

$$s_\gamma \overline{a_u}^{pr} (-\tilde{\mu} c_\gamma \overline{y_u}^{st} (-s_\gamma a_u{}^{pt} + \tilde{\mu} c_\gamma y_u{}^{pt}) (-s_\gamma a_u{}^{sr} + \tilde{\mu} c_\gamma y_u{}^{sr}) + s_\gamma \overline{a_u}^{st}$$

$$(\tilde{\mu}^2 c_\gamma{}^2 y_u{}^{pt} y_u{}^{sr} + s_\gamma a_u{}^{pt} (s_\gamma a_u{}^{sr} - \tilde{\mu} c_\gamma y_u{}^{sr})))) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}{}^{p}, m_{\tilde{q}}{}^{s}, m_{\tilde{u}}{}^{r}, m_{\tilde{u}}{}^{t}] +$$

$$\frac{3}{2} c_\gamma y_e{}^{i1i2} (\tilde{\mu}^2 c_\gamma{}^2 \overline{y_u}^{pr} \overline{y_u}^{st} (\tilde{\mu}^2 c_\gamma{}^2 y_u{}^{pt} y_u{}^{sr} + s_\gamma a_u{}^{pt} (s_\gamma a_u{}^{sr} - \tilde{\mu} c_\gamma y_u{}^{sr})) +$$

$$s_\gamma \overline{a_u}^{pr} (-\tilde{\mu} c_\gamma \overline{y_u}^{st} (-s_\gamma a_u^{pt} + \tilde{\mu} c_\gamma y_u^{pt}) (-s_\gamma a_u^{sr} + \tilde{\mu} c_\gamma y_u^{sr}) +$$
$$s_\gamma \overline{a_u}^{st} (\tilde{\mu}^2 c_\gamma^2 y_u^{pt} y_u^{sr} + s_\gamma a_u^{pt} (s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr})))))$$
$$LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] + \frac{3}{2} s_\gamma \tilde{\mu} c_\gamma^2 y_e^{i1i2} (s_\gamma^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{sr} a_d^{pt} +$$
$$\overline{a_d}^{pr} (c_\gamma^2 y_d^{sr} \overline{a_d}^{st} a_d^{pt} + \overline{y_d}^{st} (c_\gamma a_d^{pt} - s_\gamma \tilde{\mu} y_d^{pt}) (c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr})))$$
$$LF_{2,1,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] - \frac{3}{2} s_\gamma \tilde{\mu} c_\gamma^2 y_e^{i1i2} (s_\gamma^2 \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{sr} a_d^{pt} +$$
$$\overline{a_d}^{pr} (c_\gamma^2 y_d^{sr} \overline{a_d}^{st} a_d^{pt} + \overline{y_d}^{st} (c_\gamma a_d^{pt} - s_\gamma \tilde{\mu} y_d^{pt}) (c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr})))$$
$$LF_{3,1,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] + \frac{3}{2} s_\gamma \tilde{\mu} c_\gamma^2 y_e^{i1i2} (\tilde{\mu}^2 c_\gamma^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} +$$
$$\overline{a_u}^{pr} (s_\gamma^2 y_u^{sr} \overline{a_u}^{st} a_u^{pt} + \overline{y_u}^{st} (s_\gamma a_u^{pt} - \tilde{\mu} c_\gamma y_u^{pt}) (s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr})))$$
$$LF_{2,1,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] - \frac{3}{2} s_\gamma \tilde{\mu} c_\gamma^2 y_e^{i1i2} (\tilde{\mu}^2 c_\gamma^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} +$$
$$\overline{a_u}^{pr} (s_\gamma^2 y_u^{sr} \overline{a_u}^{st} a_u^{pt} + \overline{y_u}^{st} (s_\gamma a_u^{pt} - \tilde{\mu} c_\gamma y_u^{pt}) (s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr})))$$
$$LF_{3,1,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] - \frac{3}{2} s_\gamma \tilde{\mu} c_\gamma^2 y_e^{i1i2} (\tilde{\mu}^2 c_\gamma^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} +$$
$$\overline{a_u}^{pr} (s_\gamma^2 y_u^{sr} \overline{a_u}^{st} a_u^{pt} + \overline{y_u}^{st} (s_\gamma a_u^{pt} - \tilde{\mu} c_\gamma y_u^{pt}) (s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr}))) LF_{2,2,1,1,-1}[m_{\tilde{q}}^{s},$$
$$m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] - \frac{3}{2} c_\gamma y_e^{i1i2} (\tilde{\mu}^2 c_\gamma^2 \overline{y_u}^{pr} \overline{y_u}^{st} (\tilde{\mu}^2 c_\gamma^2 y_u^{pt} y_u^{sr} + s_\gamma a_u^{pt} (s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr})) +$$
$$s_\gamma \overline{a_u}^{pr} (-\tilde{\mu} c_\gamma \overline{y_u}^{st} (-s_\gamma a_u^{pt} + \tilde{\mu} c_\gamma y_u^{pt}) (-s_\gamma a_u^{sr} + \tilde{\mu} c_\gamma y_u^{sr}) + s_\gamma \overline{a_u}^{st}$$
$$(\tilde{\mu}^2 c_\gamma^2 y_u^{pt} y_u^{sr} + s_\gamma a_u^{pt} (s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr})))) LF_{2,1,1,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] +$$
$$\frac{3}{2} c_\gamma y_e^{i1i2} (\tilde{\mu}^2 c_\gamma^2 \overline{y_u}^{pr} \overline{y_u}^{st} (\tilde{\mu}^2 c_\gamma^2 y_u^{pt} y_u^{sr} + s_\gamma a_u^{pt} (s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr})) +$$
$$s_\gamma \overline{a_u}^{pr} (-\tilde{\mu} c_\gamma \overline{y_u}^{st} (-s_\gamma a_u^{pt} + \tilde{\mu} c_\gamma y_u^{pt}) (-s_\gamma a_u^{sr} + \tilde{\mu} c_\gamma y_u^{sr}) + s_\gamma \overline{a_u}^{st}$$
$$(\tilde{\mu}^2 c_\gamma^2 y_u^{pt} y_u^{sr} + s_\gamma a_u^{pt} (s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr})))) LF_{3,1,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] +$$
$$\frac{3}{4} c_\gamma y_e^{i1i2} (\tilde{\mu}^2 c_\gamma^2 \overline{y_u}^{pr} \overline{y_u}^{st} (\tilde{\mu}^2 c_\gamma^2 y_u^{pt} y_u^{sr} + s_\gamma a_u^{pt} (s_\gamma a_u^{sr} - 2 \tilde{\mu} c_\gamma y_u^{sr})) +$$
$$s_\gamma \overline{a_u}^{pr} (-2 \tilde{\mu} c_\gamma \overline{y_u}^{st} (-s_\gamma a_u^{pt} + \tilde{\mu} c_\gamma y_u^{pt}) (-s_\gamma a_u^{sr} + \tilde{\mu} c_\gamma y_u^{sr}) +$$
$$s_\gamma \overline{a_u}^{st} (\tilde{\mu}^2 c_\gamma^2 y_u^{pt} y_u^{sr} + s_\gamma a_u^{pt} (s_\gamma a_u^{sr} - 2 \tilde{\mu} c_\gamma y_u^{sr}))))$$
$$LF_{2,2,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] + \frac{3}{2} s_\gamma \tilde{\mu} c_\gamma^2 y_e^{i1i2} (\tilde{\mu}^2 c_\gamma^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} +$$
$$\overline{a_u}^{pr} (s_\gamma^2 y_u^{sr} \overline{a_u}^{st} a_u^{pt} + \overline{y_u}^{st} (s_\gamma a_u^{pt} - \tilde{\mu} c_\gamma y_u^{pt}) (s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr})))$$
$$LF_{2,1,1,1,0}[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] - \frac{3}{2} s_\gamma \tilde{\mu} c_\gamma^2 y_e^{i1i2} (\tilde{\mu}^2 c_\gamma^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} +$$
$$\overline{a_u}^{pr} (s_\gamma^2 y_u^{sr} \overline{a_u}^{st} a_u^{pt} + \overline{y_u}^{st} (s_\gamma a_u^{pt} - \tilde{\mu} c_\gamma y_u^{pt}) (s_\gamma a_u^{sr} - \tilde{\mu} c_\gamma y_u^{sr})))$$
$$LF_{3,1,1,1,-1}[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] - \frac{1}{2} c_\gamma g_1^2 g_2^2 y_e^{i1i2} LF_{2,1,1,1,-2}[\tilde{\mu}, m_1, m_2, m_{\tilde{e}}^{i2}] -$$
$$\frac{1}{2} g_1^2 g_2^2 y_e^{i1i2} (m_1 m_2 c_\gamma + s_\gamma \tilde{\mu} (m_1 + m_2) (3 c_\gamma^2 + s_\gamma^2) + 2 c_\gamma s_\gamma^2 \tilde{\mu}^2)$$
$$LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_{\tilde{e}}^{i2}] - m_1 m_2 c_\gamma g_1^2 g_2^2 s_\gamma^2 \tilde{\mu}^2 y_e^{i1i2} LF_{2,1,1,1,0}[\tilde{\mu}, m_1, m_2, m_{\tilde{e}}^{i2}] +$$
$$\frac{1}{8} c_\gamma g_1^2 g_2^2 y_e^{i1i2} LF_{2,1,1,1,-2}[\tilde{\mu}, m_1, m_2, m_{\tilde{l}}^{i1}] +$$
$$\frac{1}{8} c_\gamma g_1^2 g_2^2 y_e^{i1i2} (m_1 m_2 + 2 s_\gamma \tilde{\mu} c_\gamma (m_1 + m_2)) LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_{\tilde{l}}^{i1}] +$$
$$\frac{1}{16} c_\gamma g_1^2 y_e^{pi2} y_e^{i1r} (9 m_1 c_\gamma + 17 s_\gamma \tilde{\mu}) (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] +$$
$$\frac{1}{2} m_1 g_1^2 s_\gamma^2 \tilde{\mu}^2 y_e^{pi2} y_e^{i1r} (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) LF_{2,1,1,1,0}[\tilde{\mu}, m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] -$$
$$\frac{3}{16} c_\gamma g_1^2 \overline{y_e}^{pr} y_e^{pi2} (m_1 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma a_e^{i1r} - s_\gamma \tilde{\mu} y_e^{i1r}) LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}] +$$
$$\frac{3}{8} c_\gamma g_1^2 \overline{y_e}^{pr} y_e^{i1r} (m_1 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma a_e^{pi2} - s_\gamma \tilde{\mu} y_e^{pi2}) LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{p}] +$$
$$\frac{1}{16} c_\gamma g_2^2 y_e^{pi2} y_e^{i1r} (9 m_2 c_\gamma + 17 s_\gamma \tilde{\mu}) (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] +$$
$$\frac{1}{2} m_2 g_2^2 s_\gamma^2 \tilde{\mu}^2 y_e^{pi2} y_e^{i1r} (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) LF_{2,1,1,1,0}[\tilde{\mu}, m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] +$$
$$\frac{3}{16} c_\gamma g_2^2 \overline{y_e}^{pr} y_e^{pi2} (m_2 c_\gamma + s_\gamma \tilde{\mu}) (c_\gamma a_e^{i1r} - s_\gamma \tilde{\mu} y_e^{i1r}) LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}] +$$
$$\frac{1}{4} c_\gamma^2 \overline{y_e}^{st} y_e^{si2} y_e^{i1r} \overline{a_e}^{pr} (-c_\gamma a_e^{pt} + s_\gamma \tilde{\mu} y_e^{pt}) LF_{2,1,1,1,-1}[\tilde{\mu}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}] +$$
$$\frac{1}{4} s_\gamma \tilde{\mu} c_\gamma \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{i1t} (c_\gamma a_e^{sr} - s_\gamma \tilde{\mu} y_e^{sr}) LF_{2,1,1,1,-1}[\tilde{\mu}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}] +$$
$$\frac{1}{4} c_\gamma^2 \overline{y_e}^{st} y_e^{pi2} y_e^{i1t} \overline{a_e}^{pr} (-c_\gamma a_e^{sr} + s_\gamma \tilde{\mu} y_e^{sr}) LF_{2,1,1,1,-1}[\tilde{\mu}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}] +$$
$$\frac{1}{4} s_\gamma \tilde{\mu} c_\gamma \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{si2} y_e^{i1r} (c_\gamma a_e^{pt} - s_\gamma \tilde{\mu} y_e^{pt}) LF_{2,1,1,1,-1}[\tilde{\mu}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}] +$$
$$\frac{1}{2} g_1^2 g_2^2 (m_1 + m_2 + 2 s_\gamma \tilde{\mu} c_\gamma) (c_\gamma a_e^{i1i2} - s_\gamma \tilde{\mu} y_e^{i1i2}) LF_{1,1,1,1,1,-1}[m_1, m_2, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}, \tilde{\mu}] +$$
$$m_1 m_2 s_\gamma \tilde{\mu} c_\gamma g_1^2 g_2^2 (c_\gamma a_e^{i1i2} - s_\gamma \tilde{\mu} y_e^{i1i2}) LF_{1,1,1,1,1,0}[m_1, m_2, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}, \tilde{\mu}] +$$
$$\frac{1}{2} c_\gamma g_1^2 y_e^{pi2} (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) (c_\gamma a_e^{i1r} - s_\gamma \tilde{\mu} y_e^{i1r}) LF_{1,1,1,1,1,-1}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}, \tilde{\mu}] +$$
$$\frac{1}{2} m_1 s_\gamma \tilde{\mu} g_1^2 y_e^{pi2} (-c_\gamma \overline{a_e}^{pr} + s_\gamma \tilde{\mu} \overline{y_e}^{pr})$$
$$(-c_\gamma a_e^{i1r} + s_\gamma \tilde{\mu} y_e^{i1r}) LF_{1,1,1,1,1,0}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}, \tilde{\mu}] -$$
$$c_\gamma g_1^2 y_e^{i1r} (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) (c_\gamma a_e^{pi2} - s_\gamma \tilde{\mu} y_e^{pi2}) LF_{1,1,1,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}] -$$
$$m_1 s_\gamma \tilde{\mu} g_1^2 y_e^{i1r} (-c_\gamma \overline{a_e}^{pr} + s_\gamma \tilde{\mu} \overline{y_e}^{pr}) (-c_\gamma a_e^{pi2} + s_\gamma \tilde{\mu} y_e^{pi2})$$
$$LF_{1,1,1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}] - m_1 g_1^2 (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr})$$
$$(c_\gamma a_e^{pi2} - s_\gamma \tilde{\mu} y_e^{pi2}) (c_\gamma a_e^{i1r} - s_\gamma \tilde{\mu} y_e^{i1r}) LF_{1,1,1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] -$$
$$\frac{1}{2} c_\gamma g_2^2 y_e^{pi2} (c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr}) (c_\gamma a_e^{i1r} - s_\gamma \tilde{\mu} y_e^{i1r}) LF_{1,1,1,1,1,-1}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}, \tilde{\mu}] -$$
$$\frac{1}{2} m_2 s_\gamma \tilde{\mu} g_2^2 y_e^{pi2} (-c_\gamma \overline{a_e}^{pr} + s_\gamma \tilde{\mu} \overline{y_e}^{pr})$$
$$(-c_\gamma a_e^{i1r} + s_\gamma \tilde{\mu} y_e^{i1r}) LF_{1,1,1,1,1,0}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}, \tilde{\mu}])$$